Name:____________

Double Award Science – Biology Unit 2 Foundation Tier

Once you have completed a paper and have marked it, fill out the table below, using the grade boundaries provided to add your grade. Then list which questions you did well and which questions you struggled with. For the questions your struggled with, review why and at the reasons. The most common reasons are

- Not understanding basic concepts (e.g. simple definitions)
- Calculation errors (forgetting to square numbers, etc.)
- Not reading questions carefully
- Not answering the full question (describe vs. explain)
- Laboratory/practical setup mistakes

For these questions it is also useful add notes to the question to help you next time and take a photo of these questions. Use these then to guide further revision.

Year	Score /60	Grade	Good Questions	Bad Questions	Reasons
2017					
2018					
2019					
2022					
2023					

Grade boundaries - Raw Score

Year	G	F	E	D	C	Max UMS
2017	11	14	17	20	24	28
2018	8	12	16	20	24	28
2019	6	10	14	19	24	29
2022	11	15	19	23	28	33
2023	9	13	17	21	26	31

Surname	Centre Number	Candidate Number
Other Names		0

GCSE – NEW

3430U40-1

S18-3430U40-1

SCIENCE (Double Award)

Unit 4 – BIOLOGY 2
FOUNDATION TIER

TUESDAY, 15 MAY 2018 – AFTERNOON

1 hour 15 minutes

For Examiner's use only		
Question	**Maximum Mark**	**Mark Awarded**
1.	**8**	
2.	**4**	
3.	**9**	
4.	**7**	
5.	**11**	
6.	**6**	
7.	**7**	
8.	**8**	
Total	**60**	

3430U401
01

ADDITIONAL MATERIALS

In addition to this paper you may require a calculator and a ruler.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen. Do not use correction fluid.
Write your name, centre number and candidate number in the spaces at the top of this page.
Answer **all** questions.
Write your answers in the spaces provided in this booklet.
If you run out of space, use the additional pages at the back of the booklet, taking care to number the question(s) correctly.

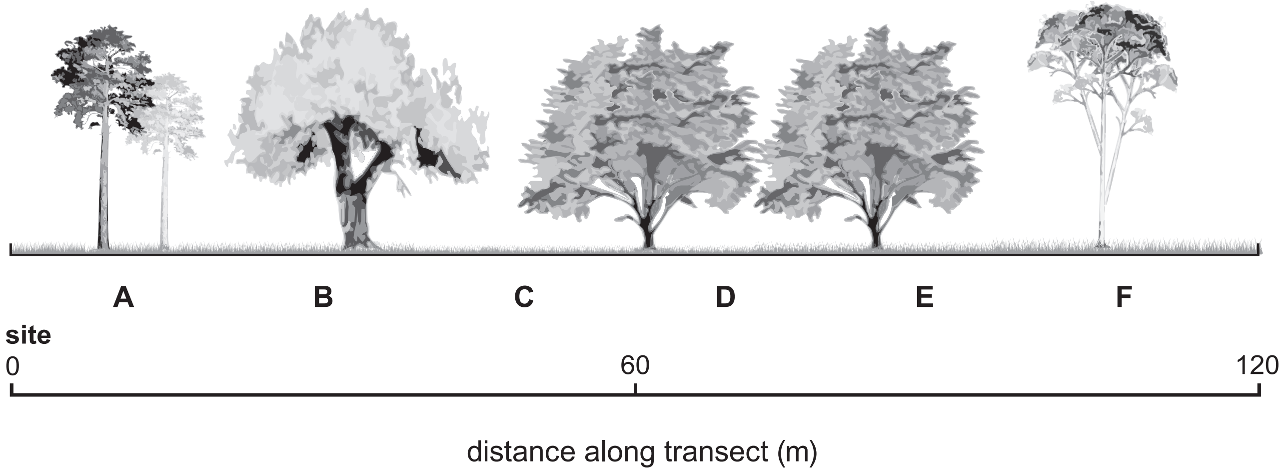
INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.
Question **6** is a quality of extended response (QER) question where your writing skills will be assessed.

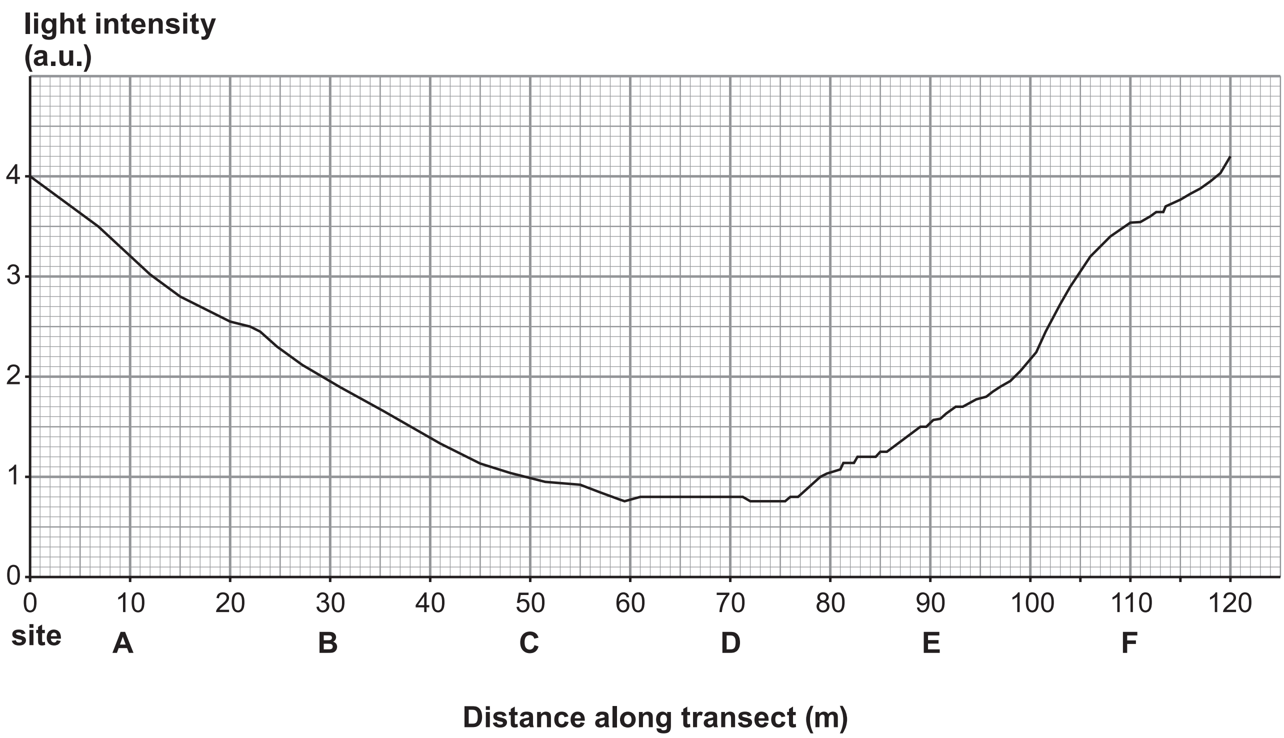
Examiner only

Answer **all** *questions.*

1. Ceri used a data logger to measure light intensity at six sites (**A** – **F**) along a 120 m transect through a wood, as shown below.



The results below are shown on her computer screen.



(i) Describe how light intensity changes along the transect. [2]

....

....

Examiner only

(ii) Ceri thought that light intensity affected the number of plant species growing under the trees along the transect.

She counted the number of plant species present at each site.

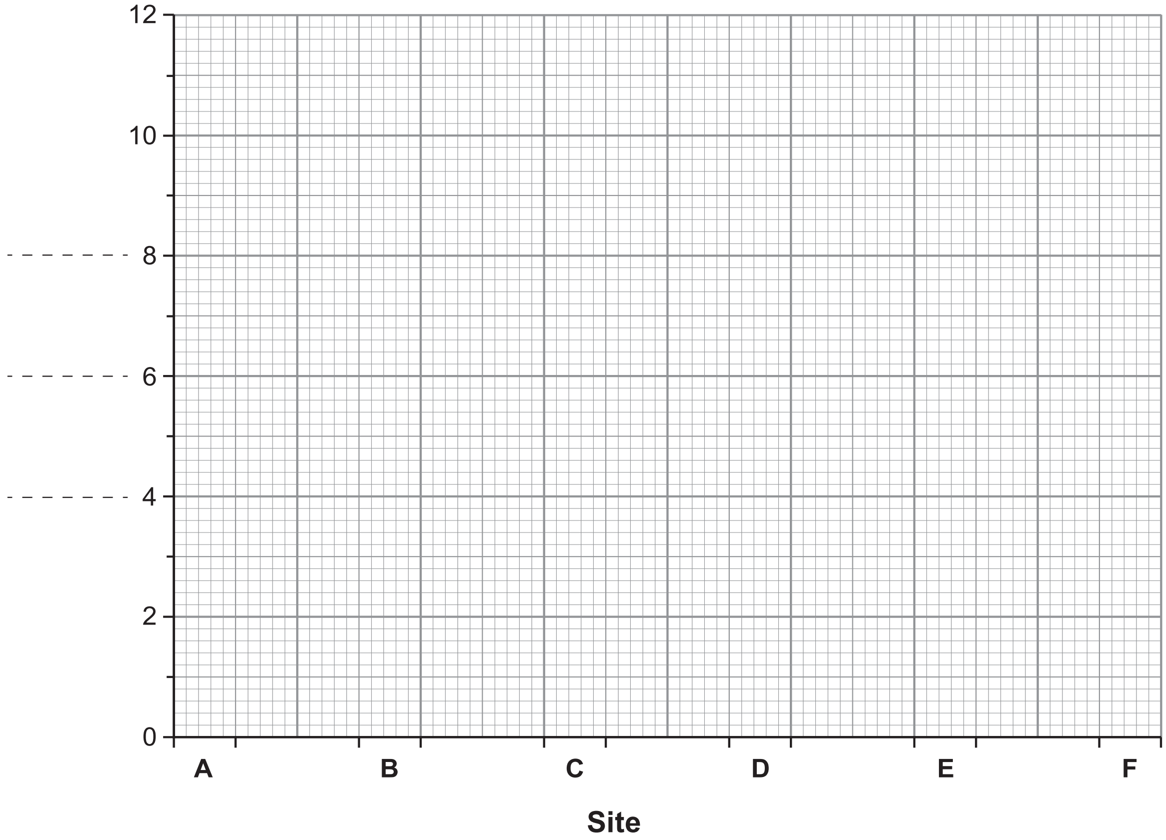
The results are shown in the table.

Site	A	B	C	D	E	F
number of plant species in 1 m^2	9	8	2	1	3	11

Complete a **bar chart** on the grid below by:

I. labelling the vertical axis; [1]

II. using a **ruler** to draw bars for the number of plant species at each site. [2]



(iii) State Ceri's hypothesis. [1]

....

....

....

3430U401 03

Examiner only

(iv) Do the results of the investigation support her hypothesis? Give a reason for your answer. [1]

....

....

(v) Apart from light intensity, suggest **one** *other* environmental factor that might affect the number of plant species growing under the trees. [1]

....

8

Examiner only

2. The photograph shows *Erinaceus europaeus*.

Erinaceus europaeus has many common names, such as hedgehog, urchin and furze-pig.

(a) State:

(i) the Kingdom to which the hedgehog belongs; [1]

....

(ii) the Genus name for the hedgehog. [1]

....

(b) Explain why scientists use the scientific name, rather than common names for organisms. [2]

....

....

....

....

3430U401
05

4

Examiner only

3. *(a)* (i) Complete the following sentences about sense organs using some of the words from the list below. [2]

impulses **signs** **stimuli** **messages** **signals**

Sense organs respond to specific .... by sending information in the form of .... which are carried to the central nervous system.

(ii) State the name of:

I. the specialised nerve cells that carry the information to the central nervous system; [1]

....

II. an organ that processes information received from sense organs. [1]

....

(b) Jed investigated the hearing of some year 11 pupils and their teachers.

He asked 12 pupils and 12 teachers to say if they could hear a buzzer when he sounded it at each of five distances away from the two groups. The same buzzer was pressed for each person.

The results are shown in the table.

Distance from buzzer (m)	Number hearing buzzer	
	Year 11 pupils	Teachers
5	12	12
10	12	12
15	12	12
20	11	7
25	8	2

Examiner only

(i) Give **one** conclusion that Jed can make from this investigation. [1]

....

....

....

....

(ii) State **one** way in which Jed's investigation is a fair test. [1]

....

(iii) What else should Jed have done to make sure the investigation was a fair test? [1]

....

(iv) Jed thought that increasing the sample size would improve his confidence in his conclusion.

Describe how Jed would increase the sample size in this investigation. Explain why it would improve his confidence in his conclusion. [2]

....

....

....

3430U401
07

9

Examiner only

4. *(a)* Complete the following sentences about genes. [2]

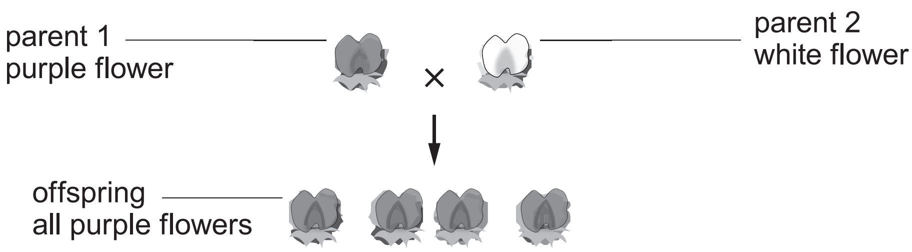
(i) Genes are found in the nucleus of a cell on paired structures called

....

(ii) Genes are sections of a long molecule called ....

(b) Alan crossed two pea plants *(Pisum sativum)*.

- Parent 1 had purple flowers.
- Parent 2 had white flowers.
- All the offspring had purple flowers, as shown below.



Flower colour in pea plants is controlled by one gene. The gene has two alleles, **D** and **d**.

(i) Underline the correct answer to complete the following sentences.

I. The genotype of parent 1 is **DD / Dd / dd** [1]

II. The genotype of parent 2 is **DD / Dd / dd** [1]

(ii) State the phenotype of the offspring. [1]

....

(iii) The offspring are heterozygous.

State the meaning of the term *heterozygous*. [2]

....

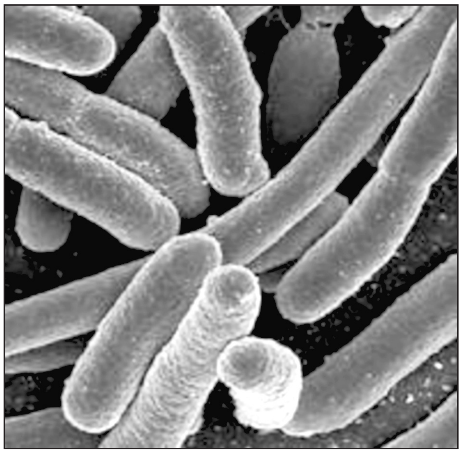
....

....

7

Examiner only

5. *(a)* The photograph shows bacteria called *E. coli.*



Bacteria such as *E. coli* can be pathogens.

(i) State the meaning of the term *pathogen*. [1]

....

(ii) Describe **two** ways in which pathogens are spread from person to person. [2]

....

....

....

....

3430U401
09

(b) Bacterial infections may be treated by antibiotics.

(i) State the name of **one** antibiotic. [1]

....

(ii) Which **one** of the following results from the over use of antibiotics?
Underline the correct answer. [1]

People become immune to antibiotics

People become resistant to antibiotics

Bacteria become immune to antibiotics

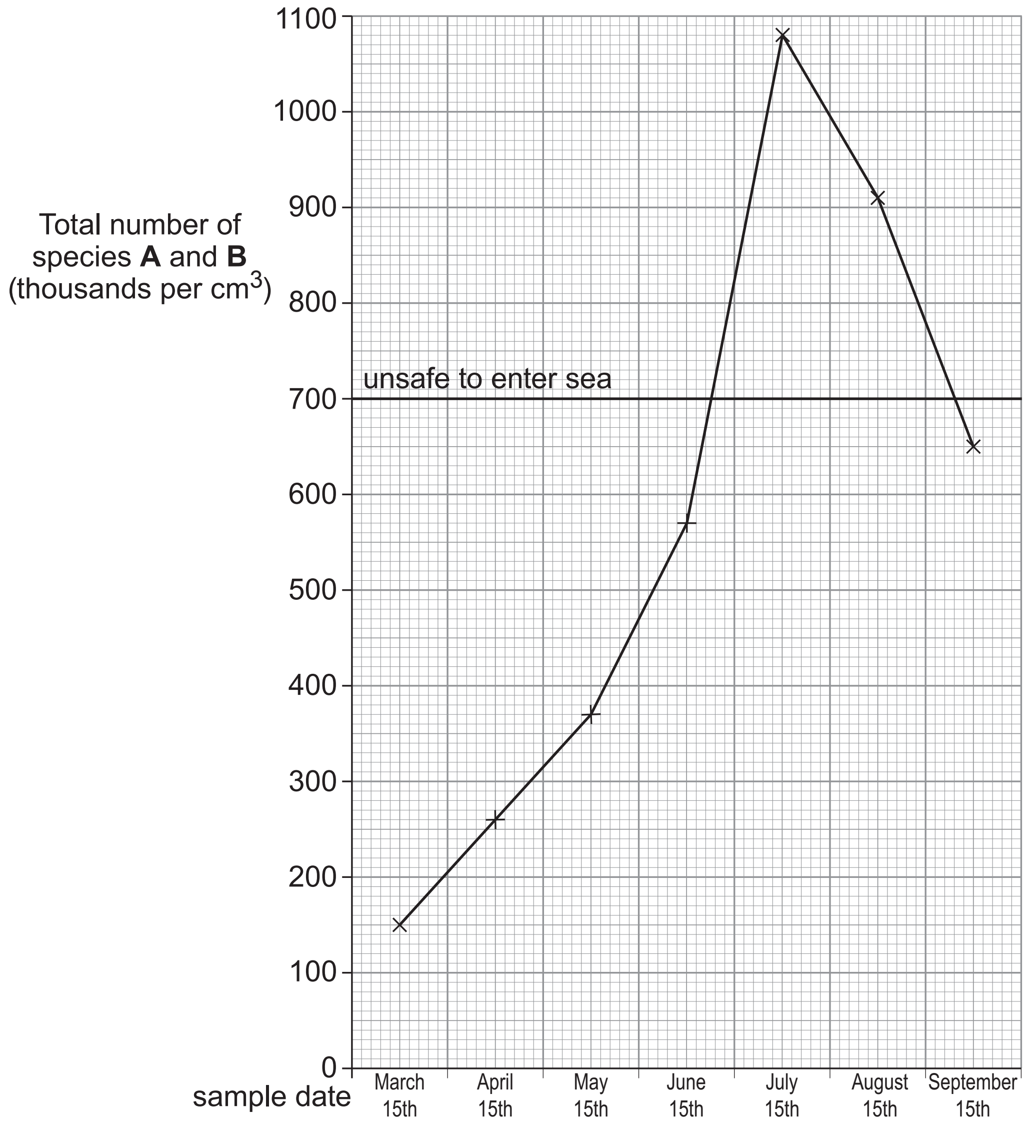
Bacteria become resistant to antibiotics

Examiner only

(c) The table shows the number of two species of bacteria (**A** and **B**) in samples of seawater taken near a beach.

Date of sample	Number of bacteria (thousands per cm^3)		Total number of species **A** and **B** (thousands per cm^3)
	Species **A**	Species **B**	
March 15th	100	50	150
April 15th	200	60	260
May 15th	300	70	370
June 15th	500	70	570
July 15th	1000	80	1080
August 15th	800	110	910
September 15th	300	350	650

(i) The graph below shows how the total number of species **A** and **B** changed between March 15th and September 15th.



Examiner only

I. Between which **two** sampling dates was the increase in the total number of species **A** and **B** the fastest? [1]

Between .... and ....

II. If the total number of species **A** and **B** in the sample is likely to be greater than 700 000 per cm^3 in any month, people are advised to stay out of the sea.

A safety officer said:

"On June 15th, I decided to advise people to stay out of the sea until further notice."

Explain why the safety officer made this decision, even though the seawater was safe on that date. [2]

....

....

....

....

(ii) I. Describe the trends shown in the **table** for the numbers of species **A** and **B**. [2]

Species **A**: ....

Species **B**: ....

II. Use your answer to part (I.) to explain why the sampling should continue into October. [1]

....

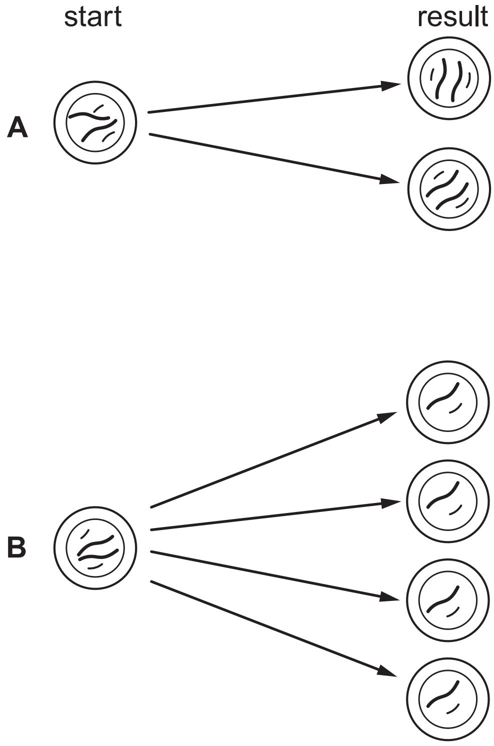
....

11

Examiner only

6. Mitosis and meiosis are two types of cell division.

Diagrams **A** and **B** show the start and result of both types of cell division in a cell with four chromosomes.



Use the diagrams and your own knowledge to describe the two types of cell division.

You should:

- state which diagram shows mitosis and which shows meiosis
- describe what you can see at the start and the result for each type of cell division
- give **one** function for each type of cell division [6 QER]

....

....

....

....

....

....

....

Examiner only

6

7. Read the following article about diabetes.

In type 1 diabetes, the cells of the pancreas that make insulin are destroyed by cells of the immune system.
You are more likely to develop it, if diabetes runs in your family.
The immune system may be triggered to act on the pancreas by a virus, pollutants, or stress causing type 1 diabetes.

In type 2 diabetes, not enough insulin is produced or cells in the liver fail to respond to the insulin that is produced. It is more likely to occur if it runs in your family, but there are also several risk factors.

These include:

- certain lifestyle choices
- certain ethnic origins
- age

Despite a rapid rise in the incidence of diabetes, there has been a 28% fall in the number of deaths from diabetes-related conditions in Wales between 2009 and 2013. This shows there has been some success in how diabetes has been managed but early diagnosis is vital.

A spokesperson for a charity promoting diabetes awareness in Wales said:

"Type 2 diabetes can be prevented, but there is no way of preventing type 1. Looking after your own health can reduce the risk of developing type 2 diabetes."

Use the above information and your own knowledge to answer the following questions.

(a) (i) Give the reason why people at high risk of developing diabetes should be tested regularly. [1]

....

....

(ii) Does the evidence given in the article support the claim made in line 18, that *"type 2 diabetes can be prevented"* in every individual? Give reasons for your answer. [1]

....

....

....

Examiner only

(b) Explain **one** way in which *"looking after your own health can reduce the risk of developing type 2 diabetes"* (lines 18 & 19). [2]

....

....

....

....

(c) Explain what would happen in the body if the *"cells in the liver fail to respond to the insulin that is produced"*. [3]

....

....

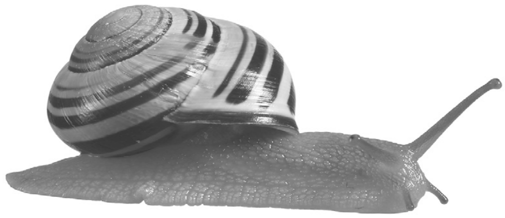
....

....

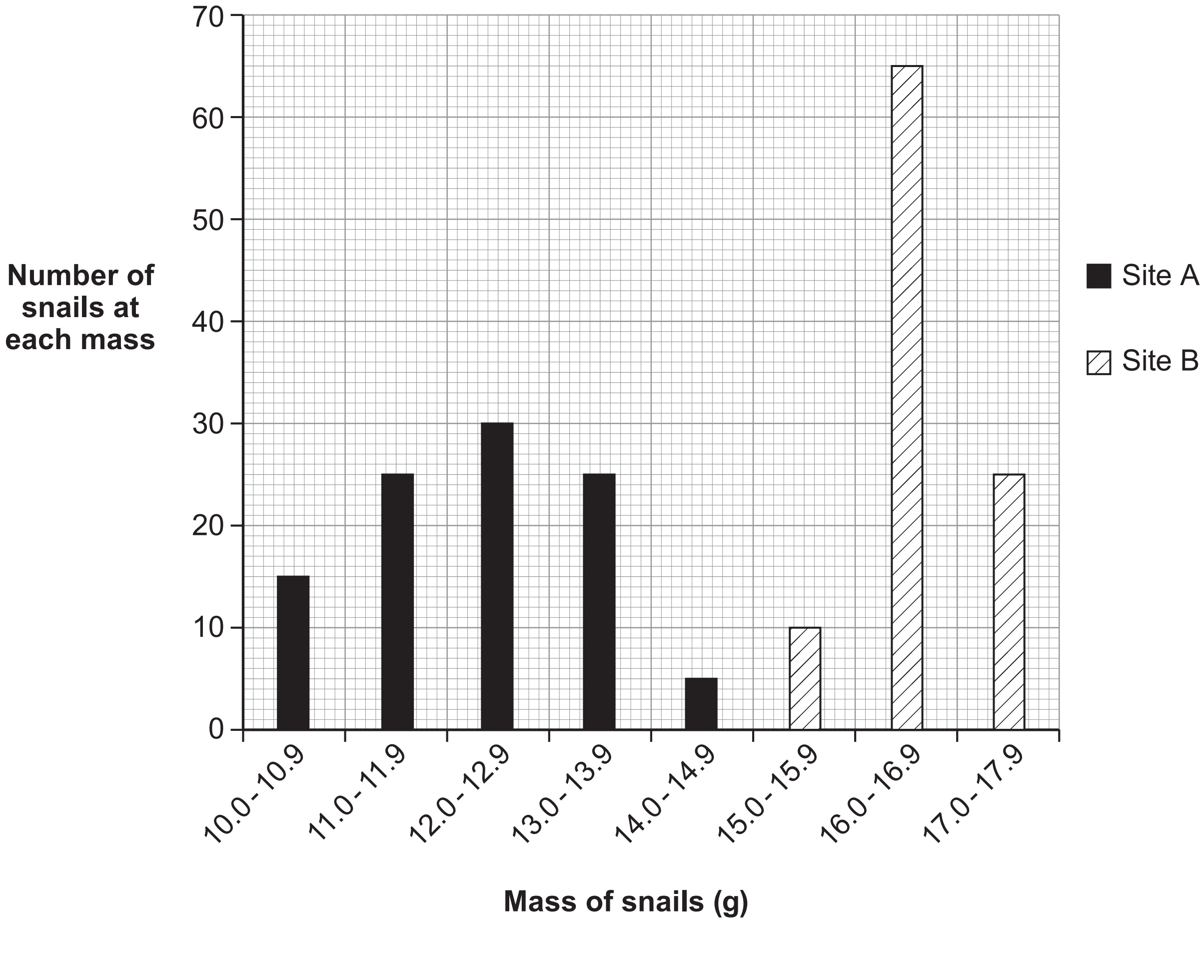
....

7

8. The photograph shows the banded snail, *Cepaea nemoralis.*



(a) Scientists investigated variation in the mass of individual snails sampled at random from two different sites, **A** and **B**. The mass of each snail was recorded to the nearest 0.1 g. The results are shown in the bar chart.



Examiner only

(i) The table below shows the mean mass of snails collected at each site.

Site	Mean mass (g)
A	12.3
B	16.8

Calculate the percentage increase in the mean mass of the snails at site **B** compared to site **A**. [2]

increase in mean mass = ... %

(ii) At which of the two sites do the snails show the greater variation in mass? Give the reason for your choice. [1]

....

....

(iii) How did the scientists reduce bias in their investigation? [1]

....

(iv) Why is it important that other scientists carry out the same investigation as these scientists? [1]

....

(b) *Cepaea nemoralis* shows genetic variation.

Use your knowledge of natural selection to explain the long term advantage of genetic variation to *Cepaea nemoralis* in a changing environment. [3]

....

....

....

....

....

....

....

8

END OF PAPER

Surname	Centre Number	Candidate Number
Other Names		0

GCSE

3430U40-1

S19-3430U40-1

SCIENCE (Double Award)

Unit 4 – BIOLOGY 2
FOUNDATION TIER

TUESDAY, 14 MAY 2019 – AFTERNOON

1 hour 15 minutes

For Examiner's use only		
Question	**Maximum Mark**	**Mark Awarded**
1.	**5**	
2.	**7**	
3.	**6**	
4.	**10**	
5.	**11**	
6.	**6**	
7.	**8**	
8.	**7**	
Total	**60**	

3430U401
01

ADDITIONAL MATERIALS

In addition to this paper you may require a calculator and a ruler.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen. Do not use correction fluid.

Write your name, centre number and candidate number in the spaces at the top of this page.

Answer **all** questions.

Write your answers in the spaces provided in this booklet.

If you run out of space, use the additional pages at the back of the booklet, taking care to number the question(s) correctly.

INFORMATION FOR CANDIDATES

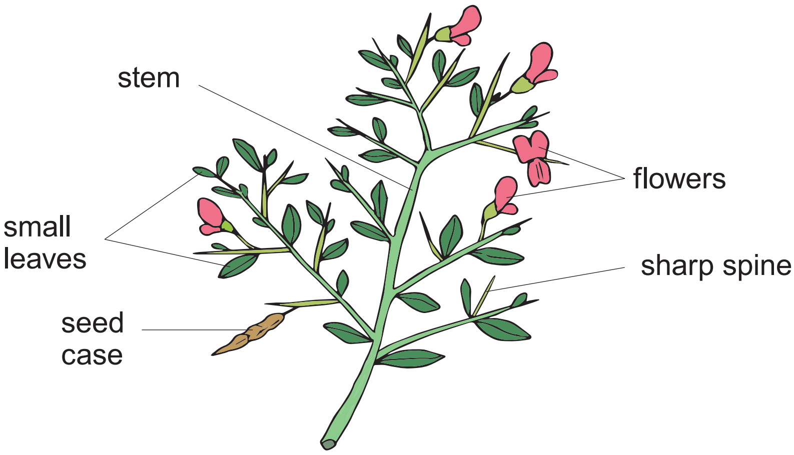
The number of marks is given in brackets at the end of each question or part-question.

Question **6** is a quality of extended response (QER) question where your writing skills will be assessed.

Examiner only

Answer **all** *questions.*

1. The diagram shows a plant called camelthorn *(Alhagi sparsifolia).*



(a) State the genus name for camelthorn. [1]

....

(b) Camelthorn lives in dry, desert areas, where there are herbivores.

Choose **two** labelled features shown on the diagram and explain how each is an adaptation for camelthorn to survive in deserts. [2]

1. ....

....

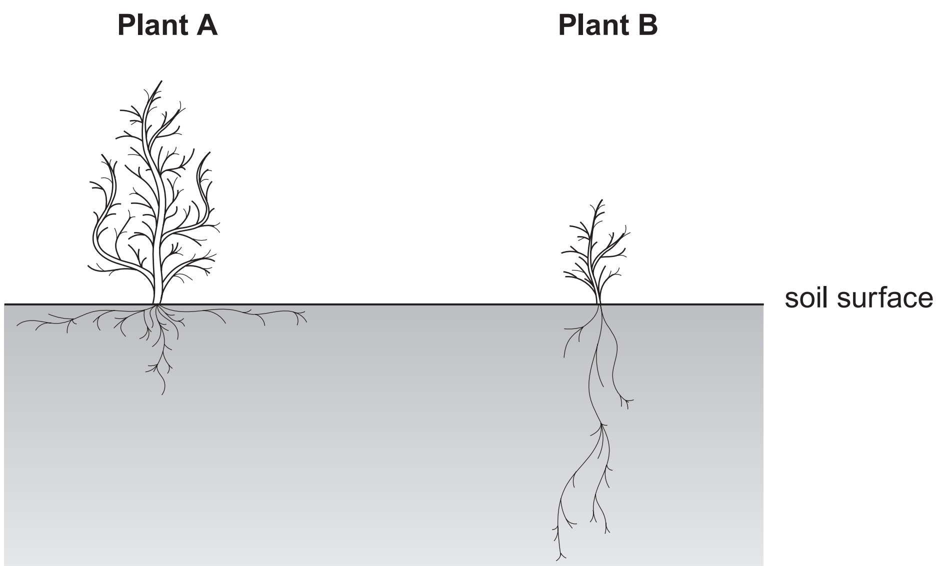
2. ....

....

Examiner only

(c) Megan investigated growth in camelthorn plants in two different conditions.

Plant A was given 250 cm^3 of water each day. **Plant B**, was watered once only at the start of the investigation. The diagram shows the appearance of the two plants at seven days.



Compare the root growth in the two plants and suggest how root growth in **plant B** is an adaptation to dry soil. [2]

....

....

....

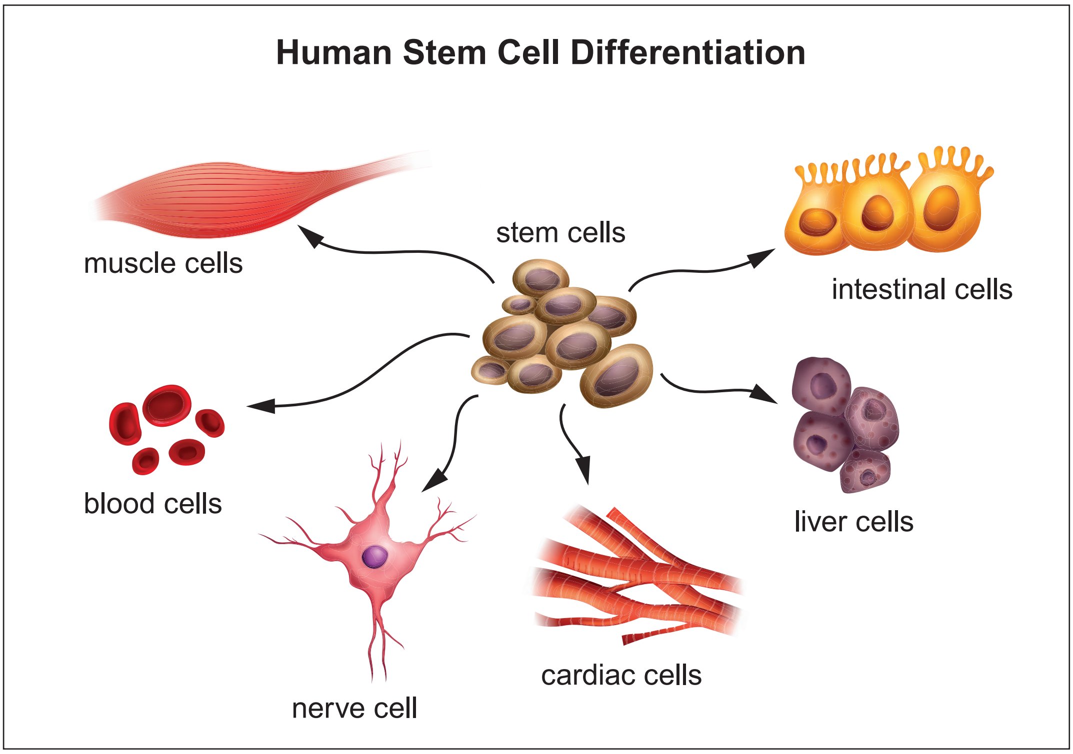
....

3430U401 03

5

Examiner only

2. The poster is about some uses of stem cells.



(a) Stem cells can differentiate.
Using **one** example from the poster, explain the meaning of differentiate. [1]

....

....

Examiner only

(b) Stem cells undergo rapid mitosis.
The table gives five features of cell division. Some features relate to mitosis and some relate to meiosis.
Complete the table by writing **true** or **false** in each row to show the features that relate only to **mitosis**. [3]

Feature of cell division	**True** or **False**
daughter cells are genetically identical	....
produces four daughter cells	....
daughter cells retain the original chromosome number	....
daughter cells have chromosomes in pairs	....
produces gametes	....

(c) A patient receiving stem cell therapy may be treated with cells taken from their own body or from embryos.
Give **three** advantages of the use of stem cells taken from a patient compared with those from an embryo. [3]

....

....

....

....

....

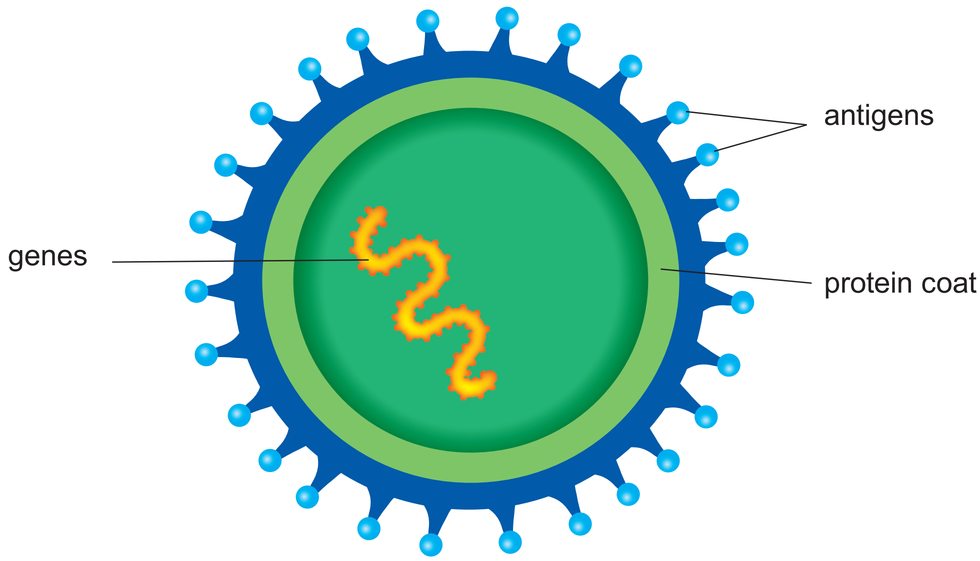
....

3430U401 05

7

Examiner only

3. *(a)* The diagram shows a pathogen that causes an infectious disease.



(i) State the type of pathogen shown above. [1]

....

(ii) Sudden changes to a gene cause changes to the antigens on the coat of the pathogen.

State the term for a sudden change to a gene. [1]

....

(b) State **one** way by which pathogens are spread between people. [1]

....

(c) The following statements (**1 – 6**) describe how the body responds when it is invaded by pathogens. The statements are not in the correct order.

1 the pathogens have foreign antigens

2 pathogens enter the blood

3 they respond by releasing antibodies to the antigens

4 the antibodies attach to the pathogen

5 the antigens are recognised as foreign by lymphocytes

6 the attached antibodies help phagocytes destroy the pathogen

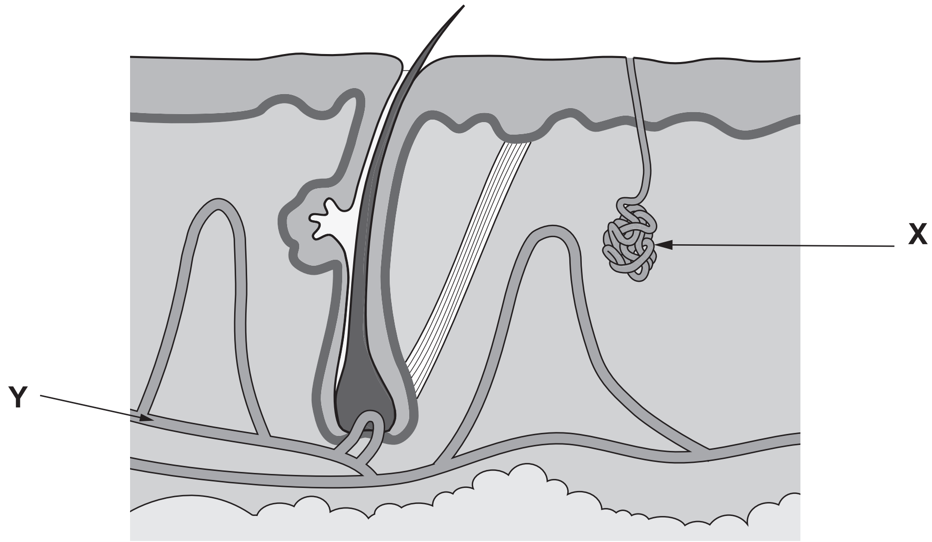
Complete the sequence to put the statements in the correct order. [3]

.... **1** .... **3** ....

6

Examiner only

4. *(a)* The diagram shows a section through the skin.

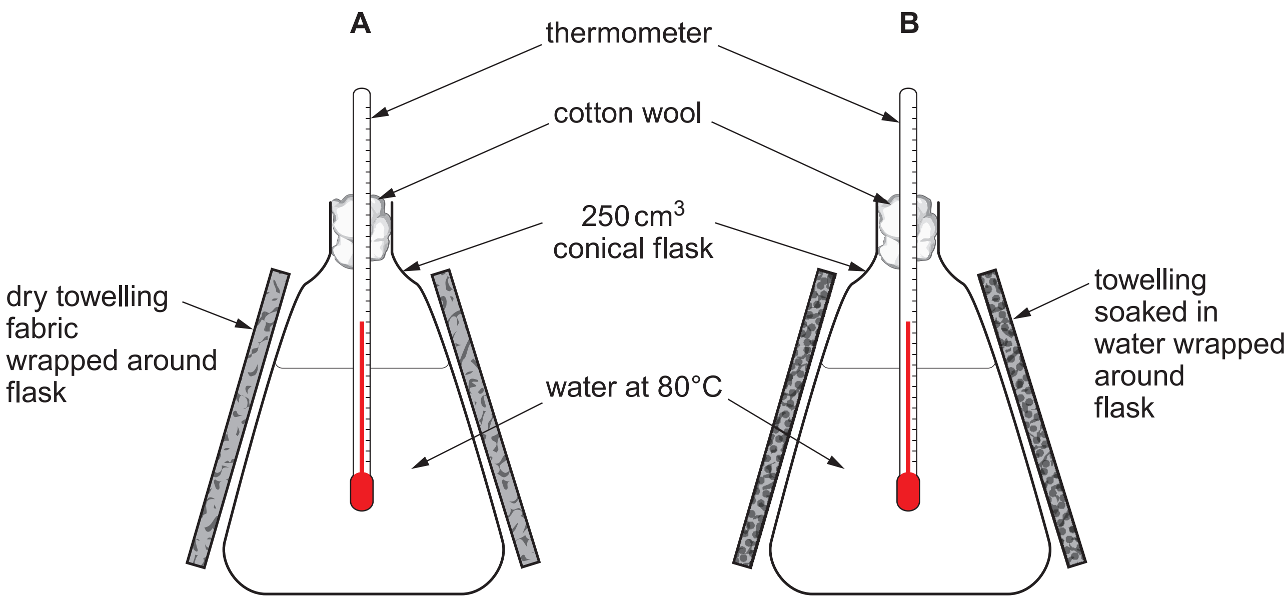


State the name of structures:

(i) **X** .... [1]

(ii) **Y** .... [1]

(b) Carwyn investigated the control of body temperature.
He set up a model to demonstrate the action of sweat using the apparatus shown below.



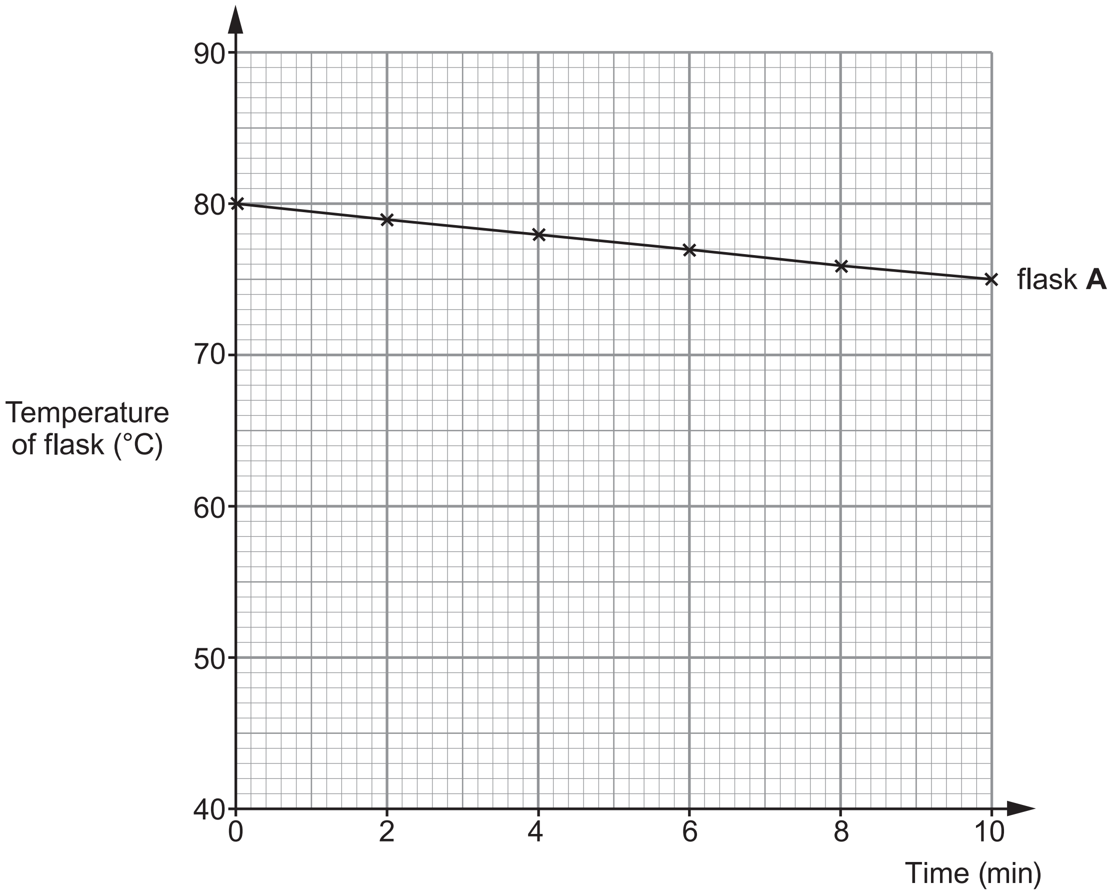
At the start of the investigation, Carwyn added 250 cm^3 of water at 80 °C to each flask. He then recorded the temperature of the water in both flasks at two minute intervals for ten minutes, as shown in the results table.

Time (minutes)	Temperature of water in flask (°C)	
	A – dry towelling	**B** – wet towelling
0	80	80
2	79	73
4	78	65
6	77	58
8	76	54
10	75	51

(i) **Complete the graph** of the results by: [3]

I. plotting the points for flask **B**

II. using a ruler to join your plots.



Examiner only

(ii) In Carwyn's model, the towelling represents the skin surface.
Using the results from flask **B**, explain the effect of sweat on human body temperature. [3]

....

....

....

....

....

....

....

(iii) Heat stroke is a condition in which body temperature becomes dangerously high.
Use the results from flask **A** to conclude why people who do not drink enough water during very hot weather may suffer heat stroke. [2]

....

....

....

....

....

10

Examiner only

5. The photograph shows an adult North Sea cod *(Gadus morhua).*

(a) Cod are important in maintaining biodiversity in the North Sea.
Choose the letter (**A - C**) from the list which defines biodiversity for an area. [1]

A The total number of species

B The variety of species and the number of organisms within those species

C The variety of species and their feeding relationships

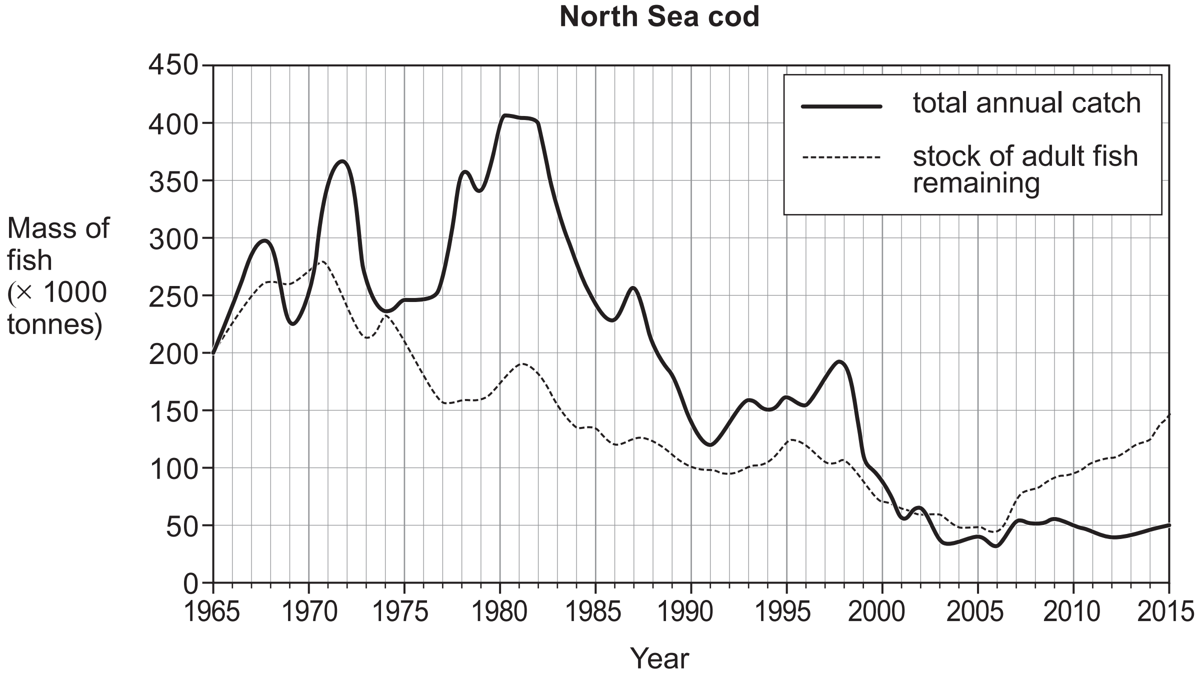
Answer ☐

(b) Read the following information about cod.

- Overfishing in the second half of the 20th century had a serious effect on the stocks of adult cod in the North Sea.
- The **total** stock of adult cod is estimated annually.
- The **safe** stock is the minimum mass of the adult cod population that should be left in the sea each year to ensure that the cod remains at sustainable levels.
- In 1996, scientists predicted that without action there would be no adult cod in the North Sea by 2015.
- In 2005, the European Commission put an annual limit (quota) on the mass of cod that could be taken by each vessel fishing in the area.

3430U401
11

The graph shows the total annual catch of cod and the stock of adult fish remaining in the North Sea from 1965 to 2015.



(i) The safe stock is 150 000 tonnes.

Use a ruler to draw a straight line on the grid to show the safe stock, from 1965 to 2015. [1]

(ii) Use the data in the graph to answer the following questions.

I. Calculate the **difference in mass** between the total catch in 1982 and the safe stock. [2]

Difference in mass = ... thousand tonnes

II. One tonne contains 120 adult cod.
Calculate the number of adult cod in the safe stock. [2]

Number of adult cod in the safe stock = ...

Examiner only

(iii) Describe the evidence that:

I. shows overfishing beyond a sustainable level took place between 1971 and 2005; [2]

....

....

....

....

II. supports the prediction made by scientists in 1996, that '*without action there would be no adult cod in the North Sea by 2015*'*;* [1]

....

....

III. shows the fishing quota of 2005 had an effect on the cod population. [1]

....

....

(c) Maintaining biodiversity ensures a continued food supply for humans.
Give **one** *other* benefit to humans of maintaining biodiversity. [1]

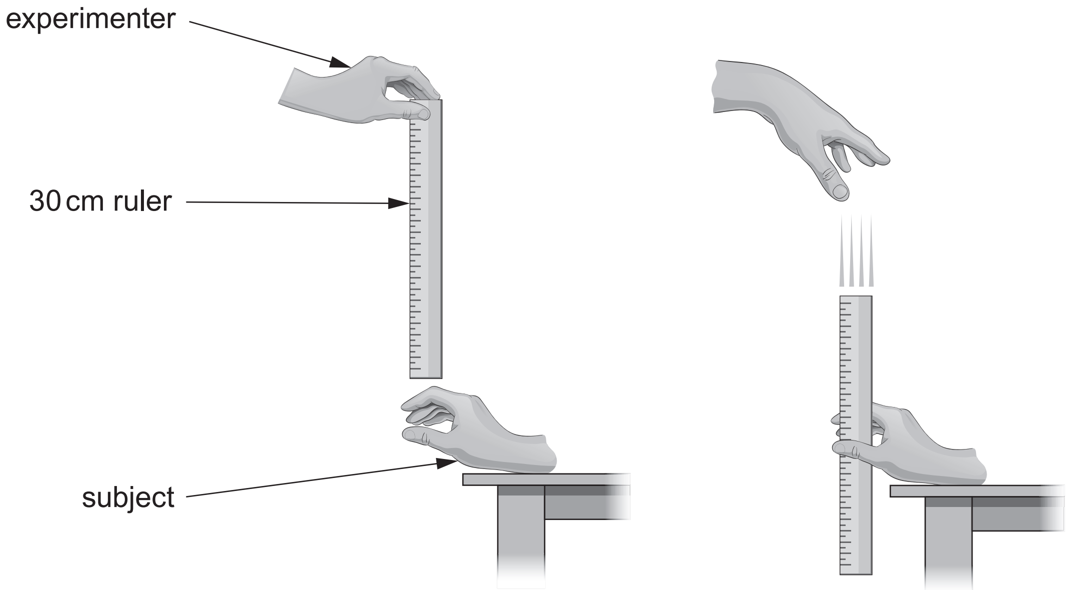
....

....

11

Examiner only

6. Reaction time can be measured using the following method.



1. The experimenter holds a 30 cm ruler just above the hand of the subject.
2. The ruler is dropped.
3. The subject then catches the ruler as quickly as possible.
4. The distance travelled by the ruler between steps 2 and 3 gives a measure of reaction time.

You have two solutions of coffee, one with caffeine and one without (decaffeinated coffee).

Describe how you would test the effect of drinking coffee with caffeine on the reaction time of a class of 20 Year 11 students. Include at least **two** variables which should be kept the same in order to make it a fair test.
[You are **not** required to write out steps 1 - 4 above in your answer.] [6 QER]

....

....

....

....

....

....

....

....

....

....

Examiner only

6

Examiner only

7. *(a)* State what is meant by the following terms:

(i) gene; [1]

....

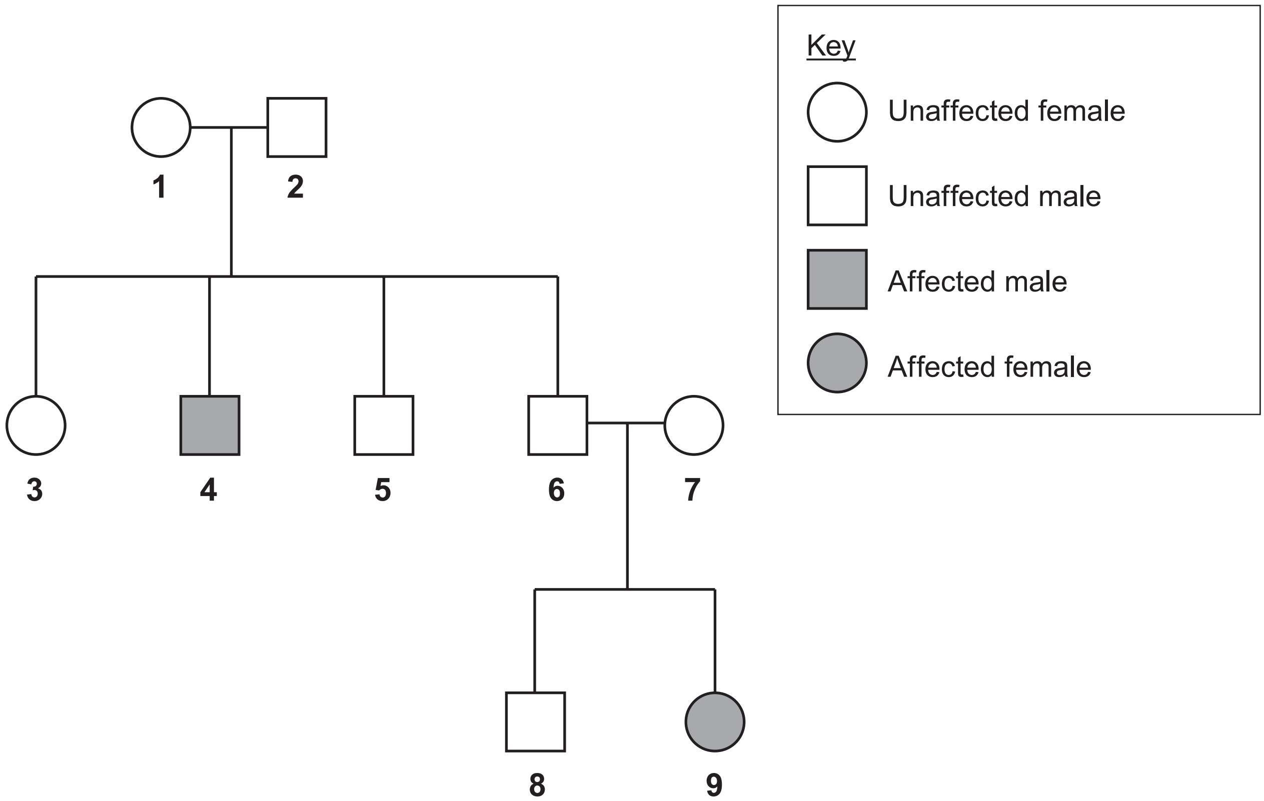
....

(ii) allele. [1]

....

....

(b) Cystic fibrosis is an inherited condition. The allele that causes this condition is recessive. The family tree below shows how cystic fibrosis is passed on in families.



Examiner only

(i) Use letters **N** and **n** to complete the Punnett square to show how parents **1** and **2** could produce offspring number **4**. [2]

Gametes		

(ii) State the genotype of individual **9**. Explain your answer. [2]

....

....

(iii) State the possible genotypes of individual **5**. [1]

....

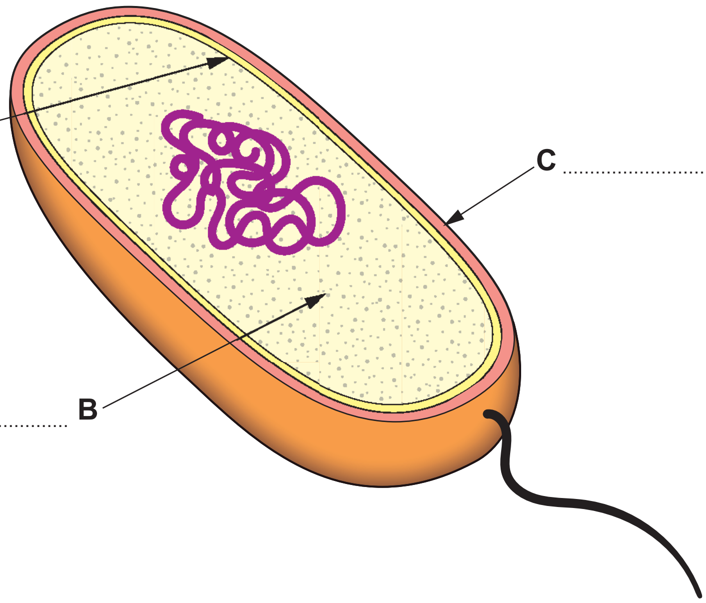
(c) Our understanding of how characteristics are inherited comes from the principles proposed by Gregor Mendel in the 19th century. In his experiments on pea plants *(Pisum sativum)* Mendel studied characteristics controlled by a single gene. State why this is not representative of most phenotypes. [1]

....

....

8

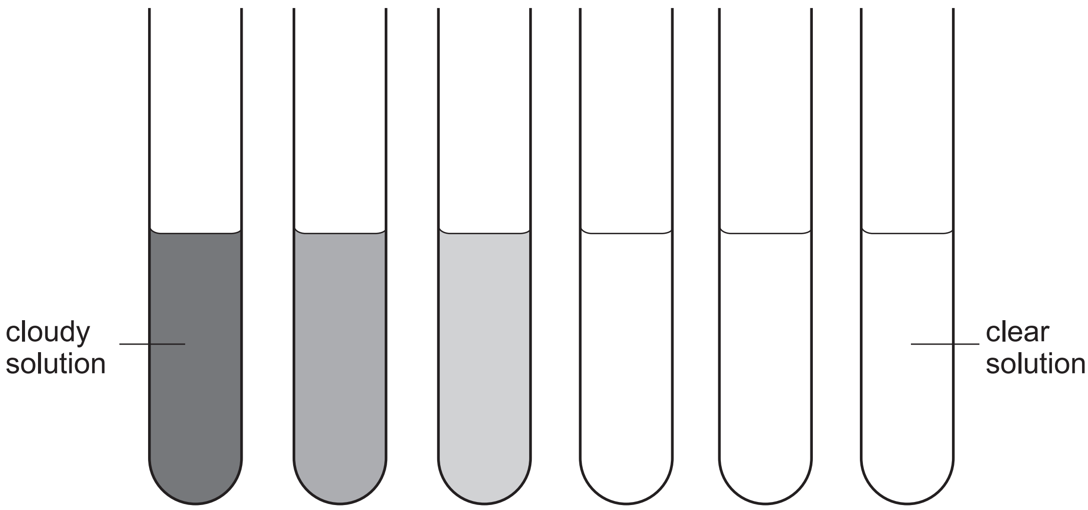
8. The diagram shows a bacterial cell.



(a) Label structures **A**, **B** and **C** on the diagram above. [3]

(b) Carys wanted to investigate the effect of an antibiotic on bacterial growth. She set up the experiment shown below with test tubes containing increasing concentrations of an antibiotic and inoculated each test tube with a fixed volume of nutrient broth containing a known concentration of bacteria.

The tubes were incubated at 25 °C for two days. The cloudier the solution becomes the more bacteria present. A clear sample means there is no bacterial growth.



Tube number	1	2	3	4	5	6
Concentration of antibiotic ($\mu g/cm^3$)	0.25	0.50	1.00	2.00	4.00	8.00

Examiner only

(i) Explain the result observed in tubes **4–6**. [1]

....

....

(ii) Suggest the minimum concentration of antibiotic that would be effective against the bacterium. [1]

minimum concentration = ... $\mu g/cm^3$

(iii) Suggest how Carys could improve her experiment so that she could obtain a more accurate value for the minimum concentration of antibiotic needed to be effective against the bacterium. [2]

....

....

....

....

7

END OF PAPER

Surname	Centre Number	Candidate Number
First name(s)		0

GCSE

3430U40-1

Z22-3430U40-1

TUESDAY, 17 MAY 2022 – MORNING

SCIENCE (Double Award)

Unit 4 – BIOLOGY 2
FOUNDATION TIER

1 hour 15 minutes

For Examiner's use only		
Question	**Maximum Mark**	**Mark Awarded**
1.	**5**	
2.	**4**	
3.	**4**	
4.	**5**	
5.	**6**	
6.	**7**	
7.	**8**	
8.	**6**	
9.	**8**	
10.	**7**	
Total	**60**	

3430U401
01

ADDITIONAL MATERIALS

In addition to this paper you may require a calculator and a ruler.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen or correction fluid.

You may use a pencil for graphs and diagrams only.

Write your name, centre number and candidate number in the spaces at the top of this page.

Answer **all** questions.

Write your answers in the spaces provided in this booklet.

If you run out of space, use the additional page(s) at the back of the booklet, taking care to number the question(s) correctly.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.

Question **8** is a quality of extended response (QER) question where your writing skills will be assessed.

JUN223430U40101

PJ*(Y21-3430U40-1)

Examiner only

Answer **all** questions.

1. (a) **Draw straight lines** to join each sense organ to the stimulus it detects.
One has been done for you. [2]

Sense Organ	**Stimulus**
nose	sound
tongue	chemicals in the air
ear	light
eye	chemicals in food and drink

(b) Complete the following sentences using words from the box. [3]

brain	**reflex**	**electrical**	**spinal cord**	**neurone**

The central nervous system is made up of the ....

and ....

The type of response shown by the body which is fast, automatic and often

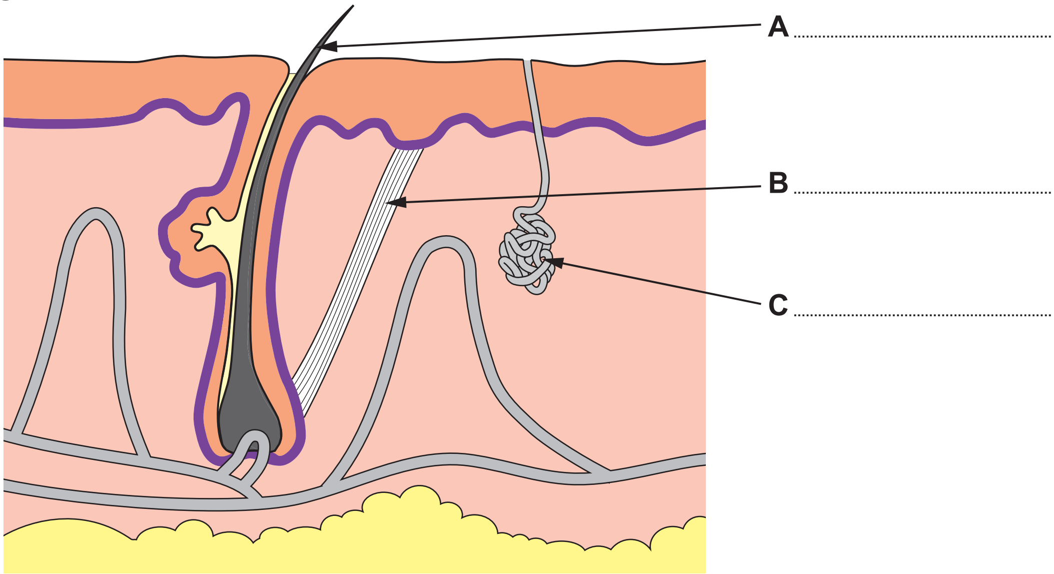
protective is called a .... action.

5

Examiner only

2. **Image 2.1** shows a section through the human skin.

Image 2.1



(a) Label parts **A**, **B** and **C** on **Image 2.1** using words from the box. [3]

sweat duct	**blood vessel**	**erector muscle**	**sweat pore**	**hair**	**sweat gland**

(b) State **one** way the skin reacts to cool us down when we get too hot. [1]

....

....

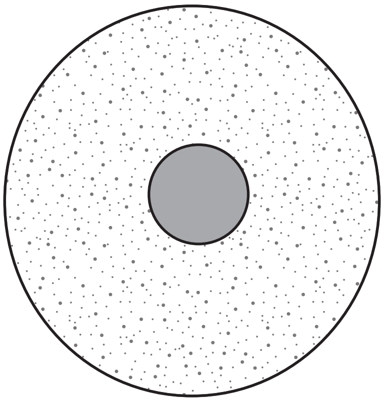
3430U401 03

4

Examiner only

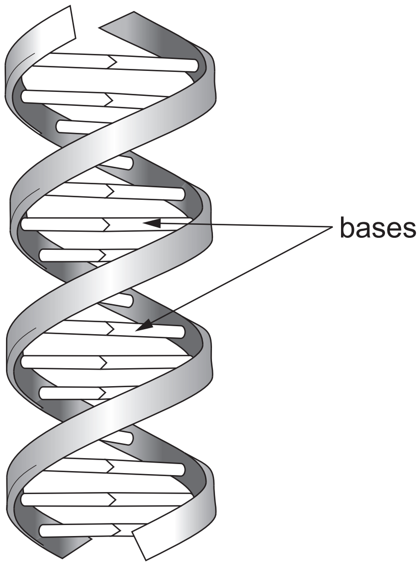
3. (a) **Image 3.1** shows an animal cell. **Draw an arrow** to show the structure where chromosomes are found. [1]

Image 3.1



Chromosomes contain DNA. **Image 3.2** shows a small section of a DNA molecule.

Image 3.2



(b) State the name used to describe the shape of the DNA molecule. [1]

....

(c) DNA contains bases that are in pairs. In the pairs of letters below, one correct pair has been underlined for you. <u>Underline</u> the other correct pair. [1]

<u>A – T</u> **G – T** **G – C** **C – A**

(d) **Complete the sentence** below by <u>underlining</u> the correct answer. [1]

The order of the bases in DNA codes for the order of amino acids in

(starch / fat / protein).

4

 (3430U40-1)

Examiner only

4. The brown rat (*Rattus norvegicus*) spreads disease. Warfarin is a poison used to kill brown rats. Over time, many brown rats have evolved to become resistant to warfarin.

(a) State the genus of the brown rat. [1]

....

(b) The brown rat is a vertebrate. State what is meant by the term vertebrate. [1]

....

(c) The sentences below describe how brown rats evolved to become warfarin resistant. They are not in the correct order:

A. the mutation made the brown rat resistant to warfarin

B. the result is an increase in the population of brown rats resistant to warfarin

C. a mutation occurred in a gene

D. brown rats that were resistant to warfarin lived long enough to reproduce

E. during reproduction the useful gene is passed on to the offspring

Use letters **A–E** from the list above to complete the sequence below to describe how brown rats became warfarin resistant. [2]

.....**C**..... .....**A**..... ....

(d) State the term used when all individuals of a species have died out. [1]

....

3430U401 05

5

Examiner only

5. MMR is a safe and effective vaccine that protects against measles, mumps and rubella. These diseases are caused by viruses.

Photograph showing a child receiving MMR vaccine

- The full course of MMR vaccination requires two doses; a child has the first injection at the age of one year and the second at three years.
- If the child comes into contact with any of these viruses in the future, lymphocytes in the blood will rapidly cause the destruction of these viruses and the child will not become ill.
- A small number of children may get minor side effects due to the vaccine.

(a) Use this information and your own knowledge to complete the following table by writing **True** or **False** against each statement. [2]

Statement	True or False
Measles is caused by a bacterium.	False
The MMR vaccine is given as drops on the tongue.	
Most children do not get side effects.	
Lymphocytes are a type of white blood cell.	

(b) Predict what would happen to the number of cases of measles, mumps and rubella if fewer children were given the MMR vaccine. [1]

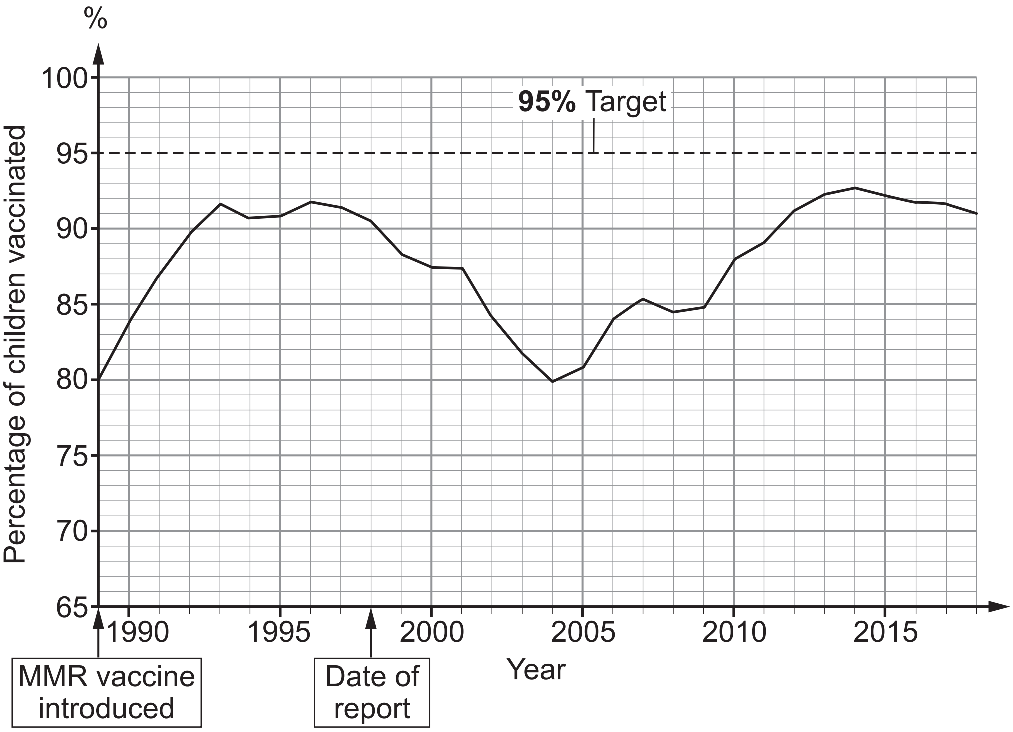
....

....

Examiner only

(c) **Graph 5.1** shows the percentage of children receiving the MMR vaccine in the UK between 1989 and 2018. It is the target of any UK vaccination programme to get at least 95% of children vaccinated.
In 1998, a report was published which claimed that there was a link between the MMR vaccine and a condition called autism. The report was later shown to be completely untrue.

Graph 5.1



(i) State the year the percentage of children receiving the MMR vaccine was closest to the UK target. [1]

....

(ii) State a conclusion that could be made about the success of the MMR vaccination programme between 1990 and 1993. [1]

....

....

....

(iii) The percentage of children vaccinated decreased after the report was published. Suggest the reason for this. [1]

....

....

6

3430U401 07

Examiner only

6. The kestrel (*Falco tinnunculus*) is a bird of prey found in Pembrokeshire, Wales. The photographs show a kestrel in flight and nesting.

Surveys are carried out by the Welsh Kite Trust to monitor the numbers of kestrels in Pembrokeshire. The data gathered by some of these surveys are shown in **Table 6.1**.

Table 6.1

Year	Number of breeding pairs
2010	12
2011	19
2012	14
2016	10

(a) (i) Calculate the percentage decrease in breeding pairs between 2011 and 2016. Use the formula below. [2]

$$\text{Percentage decrease} = \frac{\text{number of breeding pairs in 2011} - \text{number of breeding pairs in 2016}}{\text{number of breeding pairs in 2011}} \times 100$$

Percentage decrease in breeding pairs = ...

(ii) Suggest **two** possible reasons for this decrease. [2]

1. ....

....

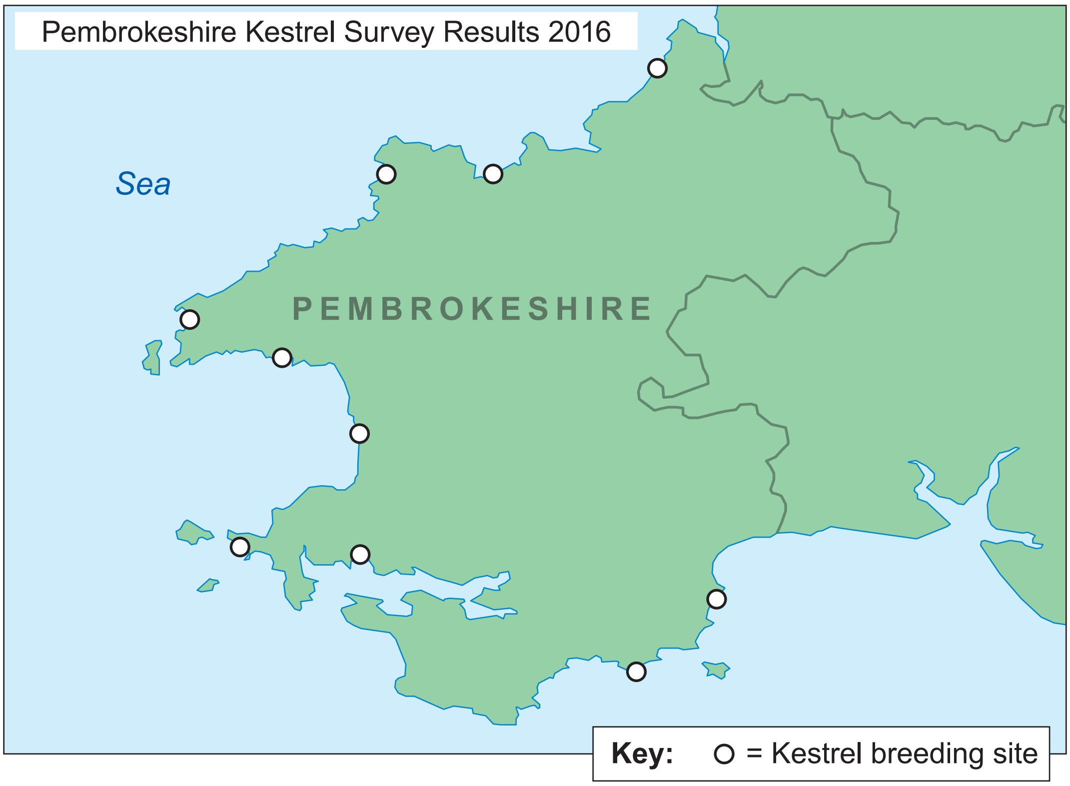
2. ....

....

Examiner only

(b) **Image 6.2** shows a map of the nesting sites of kestrels in Pembrokeshire.

Image 6.2



State **one** conclusion that can be made about where kestrels generally choose to nest. Suggest a reason for this. [2]

....

....

....

(c) Another bird of prey called the merlin (*Falco columbarius*) is closely related to the kestrel. Use your knowledge of classification to state the evidence that supports this. [1]

....

....

7

3430U401
09

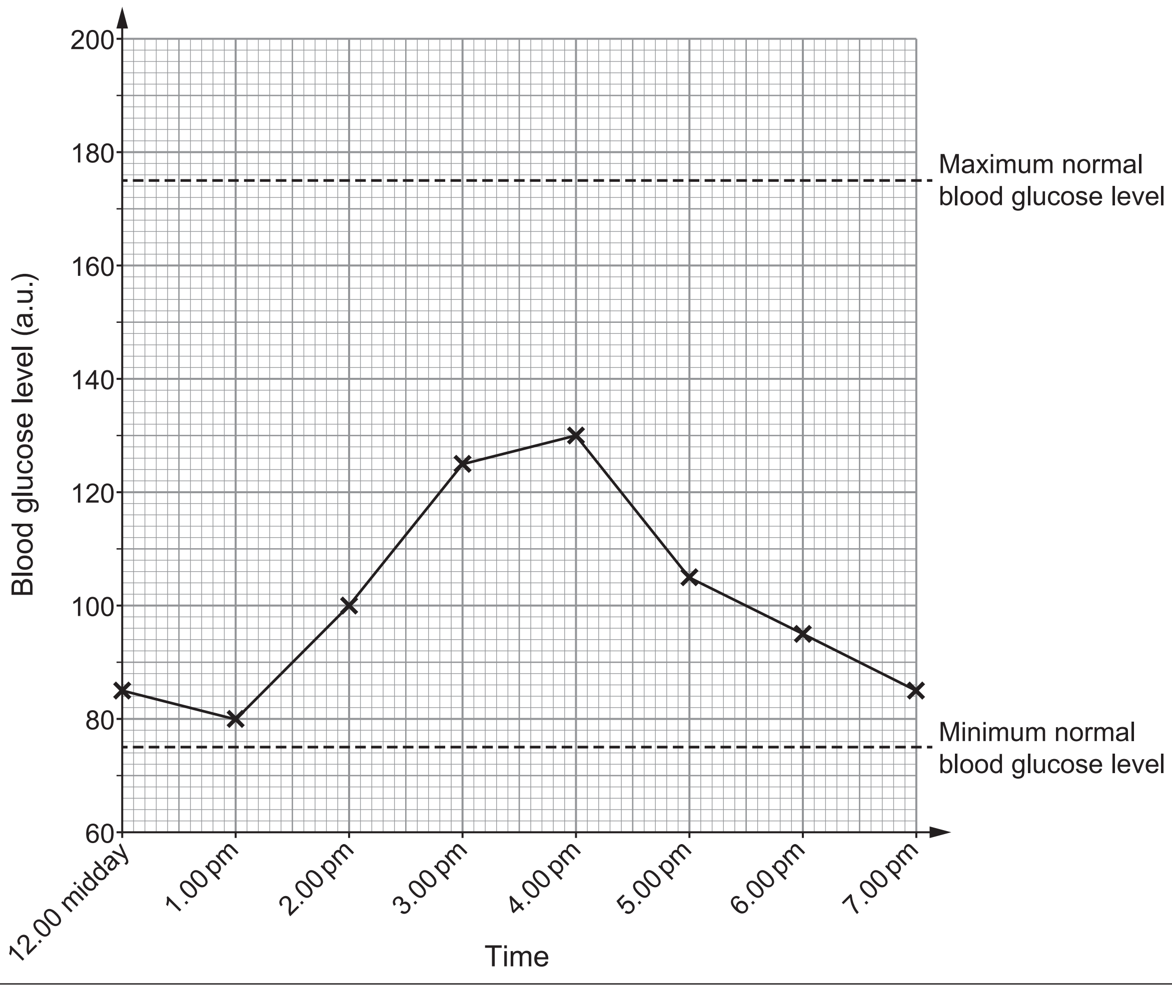
7. James eats a meal that contains glucose. A datalogger was used to record James' blood glucose levels before, during and after the meal. The data are shown in **Table 7.1**.

Table 7.1

Time	Blood glucose level (a.u.)
12.00 midday	85
1.00 pm	80
2.00 pm	100
3.00 pm	125
4.00 pm	130
5.00 pm	105
6.00 pm	95
7.00 pm	85

Graph 7.2 shows the changes in James' blood glucose level. The normal blood glucose range is also shown.

Graph 7.2



Examiner only

(a) (i) Describe the trend shown in **Graph 7.2** between 1.00 pm and 7.00 pm. [2]

....

....

....

....

(ii) State the range of James' blood glucose level. [1]

....

(iii) Suggest how **Graph 7.2** would be different in a person with diabetes. [1]

....

....

(b) (i) People with diabetes may have glucose in their urine. Describe the chemical test that could be carried out to test for the presence of glucose. [2]

....

....

....

....

(ii) State the result you would expect if James' urine was tested for the presence of glucose using this chemical test. [1]

....

(c) Glucose levels are controlled by hormones. State the name of the hormone that lowers blood glucose levels. [1]

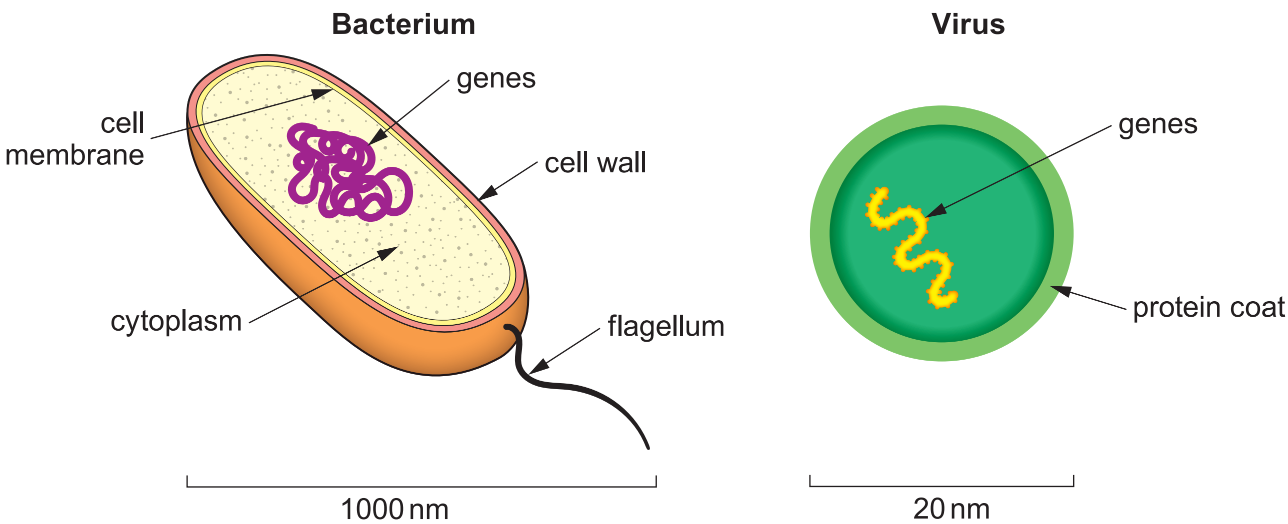
....

8

Examiner only

8. **Image 8.1** shows the structure of a bacterium and a virus.

Image 8.1



Use all the information given to describe the similarities and differences between the structure of the bacterium and the virus. [6 QER]

....

....

....

....

....

....

....

....

....

....

....

....

....

....

....

6

Examiner only

9. Cystic fibrosis is a genetic condition that affects the lungs and other organs. It is caused by a mutation.

(a) State the meaning of the term mutation. [1]

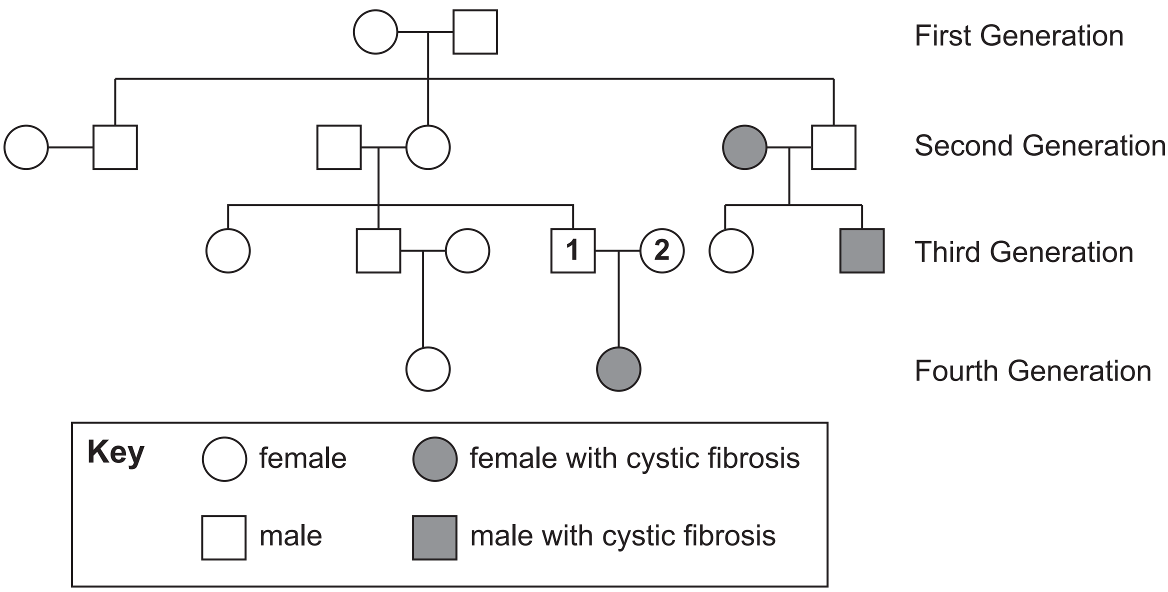
....

....

(b) State an environmental factor that will increase mutation rates. [1]

....

(c) The family tree shows the inheritance of cystic fibrosis. Cystic fibrosis is caused by a recessive allele.



Examiner only

(i) **Complete the Punnett square** below to show the possible genotypes of the future children of person **1** and person **2**. Use the letters **N** to represent the dominant allele and **n** to represent the allele that causes cystic fibrosis. [2]

(ii) Use the Punnett square to predict the probability of person **1** and person **2** having another child who has cystic fibrosis. [1]

....

(d) One treatment for cystic fibrosis is to introduce dominant alleles (**N**) into the cells lining the lungs.

(i) State the name of this type of treatment. [1]

....

(ii) State the method that is used to deliver the dominant alleles to the lungs. [1]

....

(iii) State **one** problem with the use of this method. [1]

....

....

....

8

Examiner only

10. Students used a quadrat to investigate the abundance of dandelions (*Taraxacum officinale*) on the school rugby pitch.

The method they used was:

1. Use a 1 m^2 quadrat.
2. Use a random number generator to place the quadrat on the rugby pitch.
3. Count the number of dandelions in the quadrat.
4. Repeat steps 2 and 3 another 5 times.
5. Calculate a mean.
6. Calculate the number of dandelions on the school rugby pitch.

The results they obtained are given in **Table 10.1**.

Table 10.1

Quadrat	Number of dandelions
1	3
2	5
3	2
4	7
5	15
6	6
Mean	....

Examiner only

(a) (i) **Draw a circle** around the anomalous result in **Table 10.1**. [1]

(ii) Calculate the mean number of dandelions per quadrat and **write your answer in Table 10.1**. **Do not use the anomalous result in your calculation.** [2]

(b) The teacher said that he did not have confidence in the mean. Suggest why. [1]

....

....

(c) The total area of the school rugby pitch is 7 350 m^2. Use your answer to part (a)(ii) to calculate the total number of dandelions on the school rugby pitch. [2]

Total number of dandelions on the school rugby pitch = ...

(d) State the name of the method the students should use to investigate the distribution of the dandelions across the pitch. [1]

....

7

END OF PAPER

Surname	Centre Number	Candidate Number
First name(s)		0

GCSE

3430U40-1

S23-3430U40-1

TUESDAY, 16 MAY 2023 – MORNING

SCIENCE (Double Award)
Unit 4 – BIOLOGY 2
FOUNDATION TIER

1 hour 15 minutes

For Examiner's use only		
Question	**Maximum Mark**	**Mark Awarded**
1.	**6**	
2.	**9**	
3.	**8**	
4.	**8**	
5.	**8**	
6.	**6**	
7.	**10**	
8.	**5**	
Total	**60**	

ADDITIONAL MATERIALS

In addition to this paper you may require a calculator and a ruler.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen or correction fluid.
You may use a pencil for graphs and diagrams only.
Write your name, centre number and candidate number in the spaces at the top of this page.
Answer **all** questions.
Write your answers in the spaces provided in this booklet.
If you run out of space, use the additional page(s) at the back of the booklet, taking care to number the question(s) correctly.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.
Question **6** is a quality of extended response (QER) question where your writing skills will be assessed.

JUN233430U40101

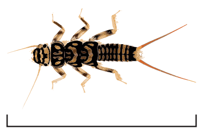
Examiner only

Answer **all** questions.

1. **Image 1.1** shows three insects:

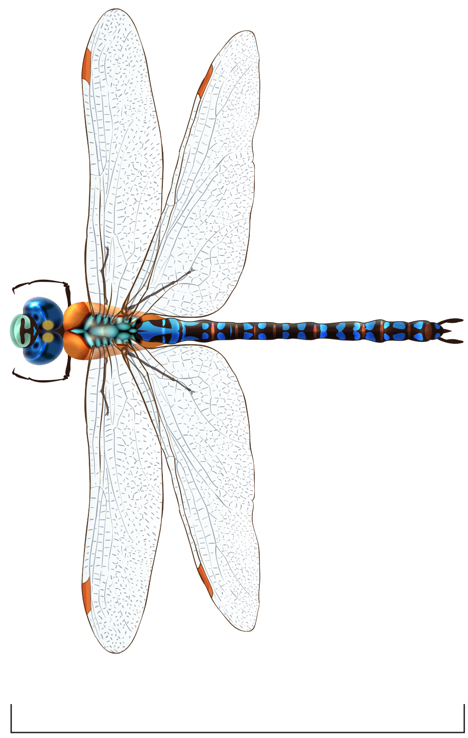
Image 1.1

Alderfly
(Sialis lutaria)
(magnification × 2)



X **Y**

Stonefly
(Diura bicaudata)
(magnification × 0.5)



Dragonfly
(*Aeshna cyanea)*
(magnification × 1)

(a) (i) Measure the line **X – Y in centimetres**. [1]

Length of the line **X – Y** = ... cm

(ii) Multiply your answer to (i) by 2 to give the actual length of the stonefly. [1]

Actual length = ... cm

Examiner only

(iii) **Complete the list below** to show the insects in order of their **actual length**. [1]

shortest Alderfly

....

longest ....

(b) **Complete Table 1.2** by writing **true** or **false** against each statement about the three insects. [3]

Table 1.2

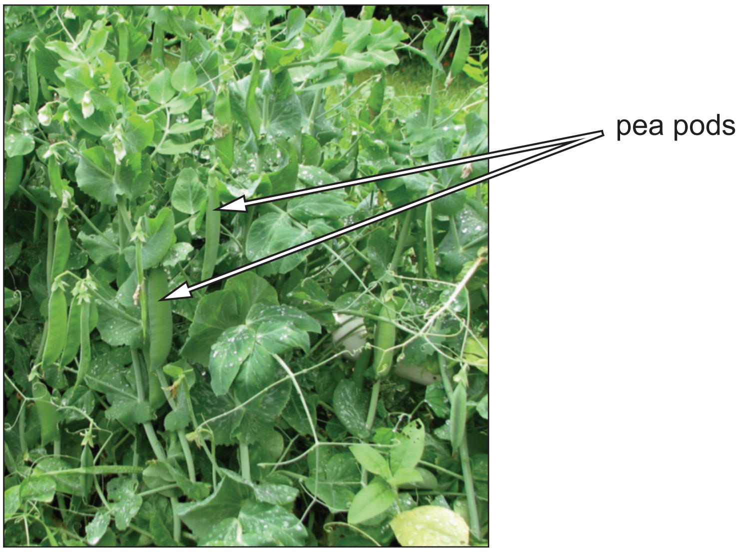
The three insects shown:	**True** or **false**
belong to the five kingdom system of classification	true
are animals	....
have backbones	....
are invertebrates	....
are in the same genus	....
each have a separate species name	....

6

3430U401 03

2. **Image 2.1** shows pea pods growing on pea plants (*Pisum sativum*).

Image 2.1



Peas grow inside the pods.

(a) Owain wanted to see if longer pods contain more peas than shorter pods.

He:

- Picked four pods from each of the two tallest pea plants on the same day.
- Measured the length of the eight pods.
- Counted and recorded the number of peas in each of the eight pods.

The results are shown in **Table 2.2**.

Table 2.2

Length of pod (cm)	Number of peas in pod
6.0	6
4.0	3
6.0	5
5.5	5
4.5	4
5.0	4
6.5	7
5.5	6

Examiner only

(i) Calculate the mean number of peas in the eight pods. [2]
Space for working.

Mean number of peas in a pod = ...

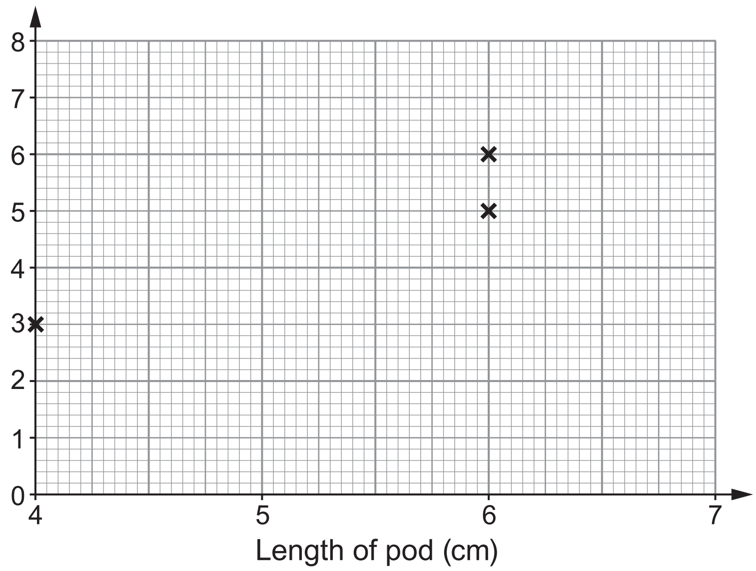
(ii) Use the results in **Table 2.2** to plot a **scattergraph** by:

I. labelling the vertical axis; [1]

II. plotting the points. The first three points have been done for you (**do not join the plots**). [2]

....

....



(iii) From the scattergraph:

state the relationship between pod length and number of peas. [1]

....

....

....

....

Examiner only

(iv) Underline **two** statements from the list to show **two** ways that Owain could increase confidence in this investigation. [2]

count the peas again

select the plants at random

measure pod length in millimetres

collect more pods from each plant

(b) Owain used two different pea plants in his investigation. They grew in two different parts of his garden.
Choose the correct letter (**A**, **B** or **C**) from the list to complete the following sentence. [1]

Variation in the number of peas in the pods is caused by:

A genetic factors only

B environmental factors only

C both genetic and environmental factors

Answer letter = ...

9

Examiner only

3. (a) Geraint is reading a book.

Complete the following sentences by choosing the correct words from the list below. [3]

signals **receptors** **impulses** **messages** **neurones**

Geraint's eyes have .... that detect light reflected from the book.

Information about the light is turned into .... which travel to the brain

along nerve cells called ....

(b) Scientists investigated the effect of drinking alcohol on the speed and accuracy of reading a paragraph of text.

The scientists took 24 men and split them into two groups, **A** and **B**.

- The men in group **A** each drank 500 cm^3 of beer containing 5% alcohol.
- The men in group **B** each drank 500 cm^3 of non-alcoholic beer.
- Both groups were then given a piece of text to read out loud.
- The time taken to read the text and the number of errors made were recorded.
- The means for each group were then calculated.

The results are shown in **Table 3**.

Table 3

Group	Mean time to complete the reading (s)	Mean number of errors
A	65	3
B	58	0

(i) State **two** conclusions that you can make from the results about the effect of alcohol on reading. [2]

Conclusion 1 ....

....

Conclusion 2 ....

....

Examiner only

(ii) State **two** ways in which the method shown made sure the men in group **A** each drank the same amount of alcohol. [2]

1. ....

....

2. ....

....

(iii) Group **B** is the control. Explain the importance of the control group in this investigation. [1]

....

....

3430U401
09

8

Examiner only

4. (a) **Choose words from the list below** to complete the following sentences.

ATP cytoplasm DNA nucleus protein vacuole

(i) Chromosomes are found in a cell structure called the .... [1]

(ii) Genes are sections of a long molecule with a double helix shape called

.... [1]

(b) A grower crosses two plants, each with pink flowers.

×

The number of offspring plants is shown in **Table 4.1**. Some of the offspring had white flowers.

Table 4.1

Flower colour	Number of offspring
pink	21
white	7

(i) Calculate the ratio of pink flowers to white flowers. [1]

ratio of pink flowers to white flowers = ... : ....

Examiner only

(ii) Complete the **two** sentences below by underlining the correct term for each.

I. The results of the cross show that the genotype of the parents is: [1]

homozygous dominant

homozygous recessive

heterozygous

II. The genotype of the offspring with white flowers is: [1]

homozygous dominant

homozygous recessive

heterozygous

(c) (i) In humans, the X and Y chromosomes are known as the sex chromosomes. Complete **Table 4.2** by placing **one** tick (✓) in each row to show the sex chromosome that may be present in the gametes. [2]

Table 4.2

Sex	Sex chromosome that may be present in the gametes		
	X only	Y only	X or Y
female			
male			

3430U401
11

(ii) Jon and Catrin have two daughters. Catrin is expecting their third child. State the probability that this child will be a boy. [1]

Probability = ...

8

Examiner only

5. **Image 5** shows surgery on a patient with heart disease caused by a faulty gene.

Image 5

(a) After surgery, it is important that wounds are sealed.
Explain why wounds must be sealed. [2]

....

....

....

....

(b) Patients are given antibiotics after surgery.
State the name of **one** antibiotic **and** the reason why antibiotics are given to patients. [2]

Antibiotic ....

Reason ....

....

(c) The surgeons are using stem cells taken from the patient to replace damaged heart muscle.
Suggest **one** benefit of using stem cells from the patient rather than from a donor. [1]

....

(d) Scientific research has produced a map of all the genes present in humans.

(i) Complete the sentence below by underlining the correct term. [1]

The complete map of human genes is called the human:

genome **gene pool** **genotype**

Examiner only

(ii) Suggest **two** ways by which scientists make their work known to other scientists around the world. [2]

I. ....

....

II. ....

....

8

Examiner only

6. Species **A** and **B** shown in **Image 6.1** are common moorland plants.

Image 6.1

Species **A**

Species **B**

Image 6.2 shows moorland in Wales.

Image 6.2

Plant distribution can be studied using the following equipment:

- 50 m tape measure
- a 1 m^2 quadrat
- notebook
- pencil
- ruler
- graph paper.

Using **all** the above equipment, describe how you would:
- investigate the distribution of species **A** and **B** between points **X** and **Y** in **Image 6.2**, which are 50 m apart
- display your results. [6 QER]

....

....

....

....

....

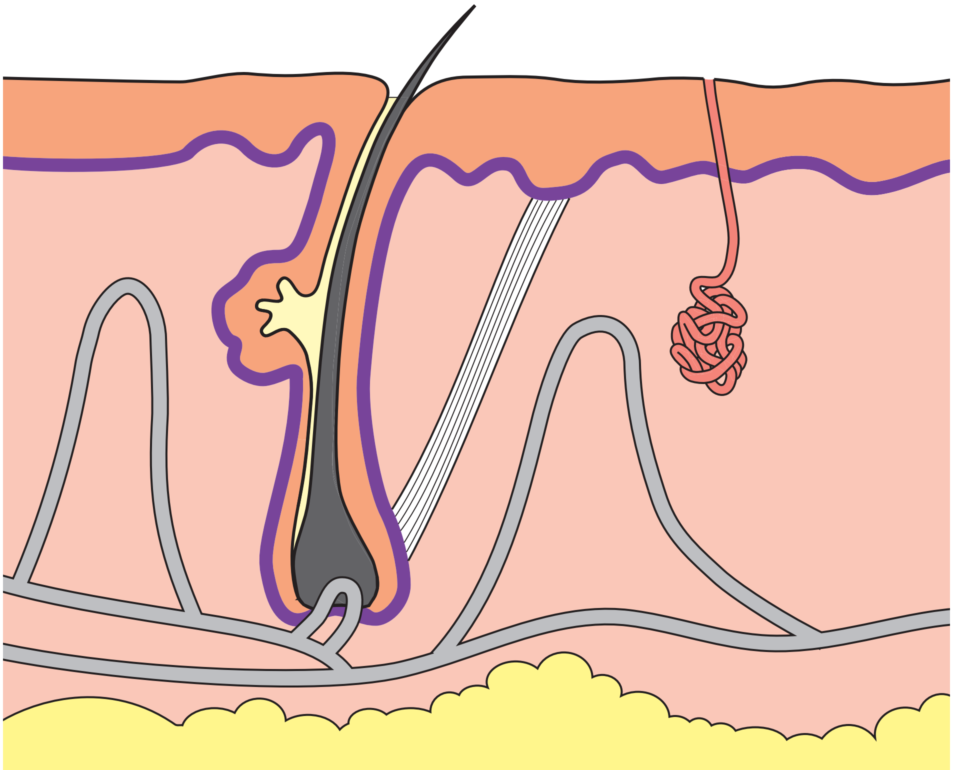
Examiner only

6

Examiner only

7. **Image 7.1** shows a section through human skin.

Image 7.1



(a) On **Image 7.1**, **use arrows to label**: [2]

I. the sweat duct

II. a blood vessel

Examiner only

(b) Scientists investigated the effect of air temperature on both the skin temperature and core body temperatures of five volunteers. Core body temperature is the temperature of the internal organs of the body.

- The five volunteers were placed in a temperature controlled laboratory at −20 °C.
- The volunteers wore bathing suits and were kept at the temperature for 5 minutes after which their skin and core temperatures were recorded.
- The experiment was then repeated at other air temperatures over the next six days, each day at a different air temperature.

The skin temperatures at different air temperatures are shown in **Table 7.2**. The mean core body temperatures are shown on **Graph 7.3**.

Table 7.2

Air temperature (°C)	Skin temperature of volunteers (°C)					Mean skin temperature (°C)
	Volunteer number					
	1	2	3	4	5	
−20	21.4	22.0	22.6	21.8	21.5	21.9
−10	23.3	23.2	22.8	22.9	23.0	23.0
0	23.8	24.2	24.4	23.5	23.9	24.0
10	25.7	25.3	25.9	25.2	25.2	25.5
20	28.3	28.1	28.5	27.9	27.8	28.1
30	33.0	32.3	32.7	32.4	32.2	32.5
40	38.5	40.2	39.3	40.2	39.1	....

(i) **Complete Table 7.2** by calculating the mean skin temperature of the five volunteers at a temperature of **40 °C**. [1]

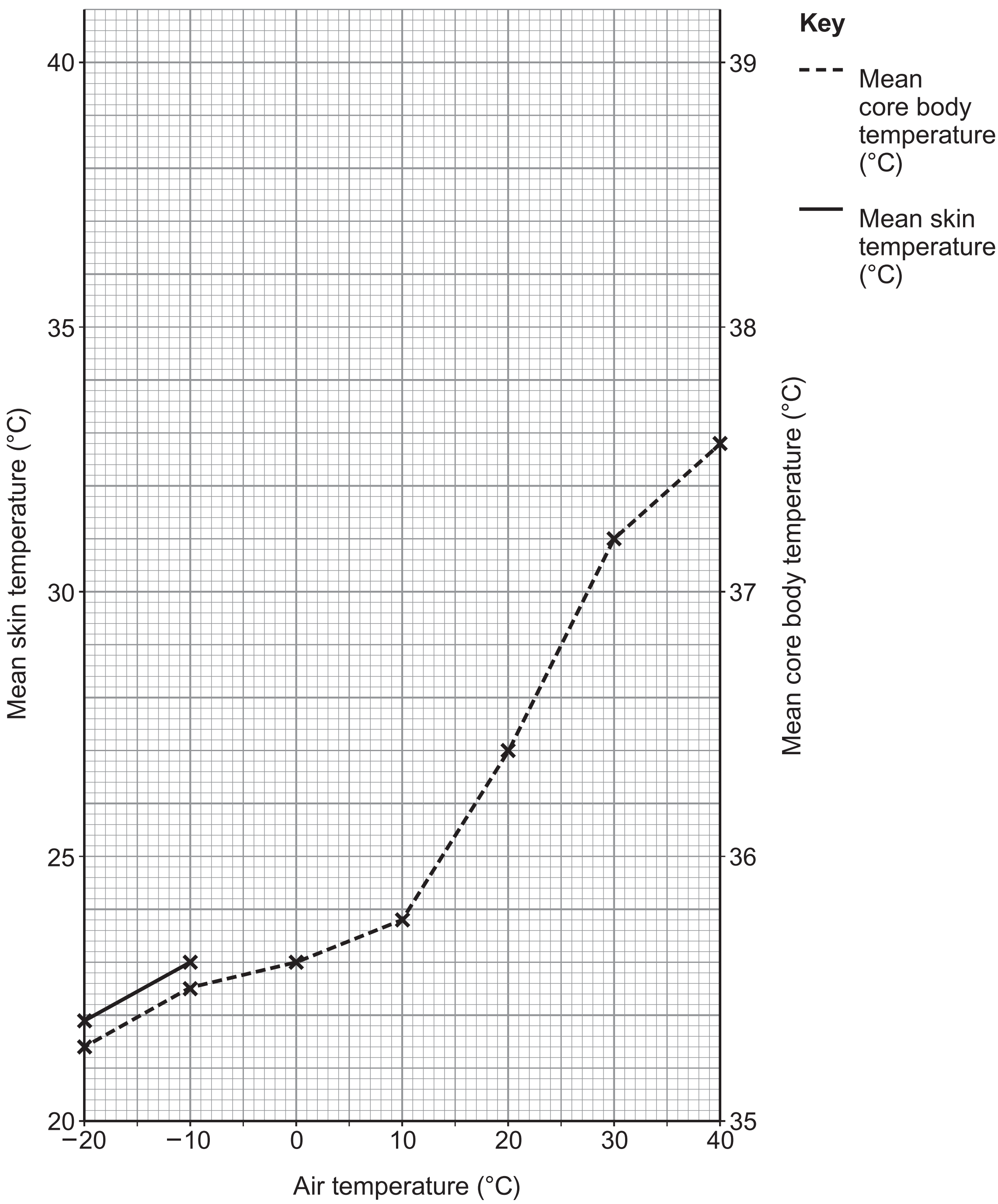
(ii) I. Plot the **mean skin temperature** against the **left-hand** y-axis on **Graph 7.3**. The first two points have been plotted for you. [2]

II. **Join the plots with a ruler.** [1]

The mean core body temperature is already plotted against the right-hand y-axis.

Examiner only

Graph 7.3



III. **From Graph 7.3** determine the mean core body temperature when the air temperature **is 25 °C**. [1]

Mean core body temperature when air temperature is 25 °C = ... °C

IV. From **Graph 7.3** state the relationship between the skin temperature and the core body temperature. [1]

....

....

Examiner only

(c) (i) Place a tick (✓) in **one** of the boxes below to show the processes which occur in the skin when air temperature is **low**. [1]

- [ ] blood vessels constrict, sweating reduces, hairs lowered on skin surface, shivering occurs
- [ ] blood vessels dilate, sweating reduces, hairs raised on skin surface, shivering occurs
- [ ] blood vessels constrict, sweating reduces, hairs raised on skin surface, shivering occurs
- [ ] blood vessels constrict, sweating increases, hairs raised on skin surface, shivering occurs

(ii) Place a tick (✓) in **one** of the boxes below to show the processes which occur in the skin when air temperature is **high**. [1]

- [ ] blood vessels dilate, sweating increases, hairs lowered on skin surface, shivering occurs
- [ ] blood vessels dilate, sweating increases, hairs lowered on skin surface, no shivering
- [ ] blood vessels constrict, sweating increases, hairs lowered on skin surface, no shivering
- [ ] blood vessels dilate, sweating decreases, hairs lowered on skin surface, no shivering

10

Examiner only

8. (a) State the meaning of the term biological control. [1]

....

....

(b) Many species of moths destroy food crops. The global effect of these pest species results in the loss of many billions of dollars per year. The stage in the life cycle of moths that causes the damage is the caterpillar, which eats the plant crop.

The simplified life cycle of most moths is shown in **Image 8.1**:

Image 8.1

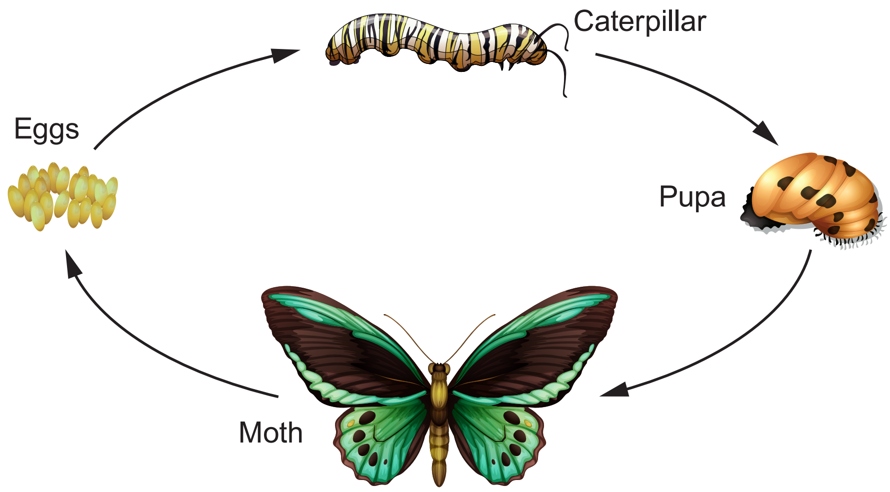


Table 8.2 shows five species of small wasps (Trichogramma) used as biological control agents. It also shows the stage in the moth life cycle they attack and the area of farmland in different countries where these biological control agents are already used successfully.

Table 8.2

Species of wasp	Stage of moth life cycle attacked by wasp	Food crops damaged by moth caterpillars	Region where wasp is used	Area of farmland where the wasp is used successfully (million hectares)
Trichogramma pretiosum	eggs	potatoes	China	2.0
Trichogramma platneri	caterpillars	corn, sugar cane	Mexico	1.5
Trichogramma acacioi	caterpillars	avocado and other fruit crops	Brazil	1.0
Trichogramma japonicum	caterpillars	rice, potatoes	South East Asia	0.3
Trichogramma ostriniae	caterpillars	corn	Europe	0.1

Examiner only

As an advisor for the United Nations you are asked to advise the Government of Peru on the use of a biological control agent to prevent moth damage in potato crops.

(i) Using all the information given, suggest the best species of *Trichogramma* to use and give the reason for your choice. [2]

....

....

....

....

(ii) State **one** factor that would need to be investigated before *Trichogramma* is released into the environment in Peru. [1]

....

....

(iii) Suggest why it is not possible to conclude, from the information in the table, that the percentage of crops damaged by moths is less in China than it is in Europe. [1]

....

....

5

END OF PAPER